A Constructive Renormalization Program for 4D SU(2) Yang–Mills (OS Reconstruction, Mass Gap, Ward Identities, Loop Equation, and Coupling Pinning)

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Abstract

This paper unifies a constructive program for four-dimensional lattice SU(2) gauge theory along a scaling trajectory $\beta = \beta(a)$ as the lattice spacing $a \downarrow 0$. The core analytic engine is a strict block contraction for a renormalization-group (RG) transfer operator obtained by integrating interior variables of a 2×2 tile. From this contraction we deduce exponential clustering and a spectral mass gap, pass Osterwalder–Schrader (OS) axioms to the continuum, identify a nontrivial renormalized local observable via the spin-1 character channel, and establish gauge-invariant Ward identities in the scaling limit. As a final gauge-invariant signature we derive the continuum loop equation and provide a tiny-loop anchor asymptotic that pins the renormalized coupling, isolating uniqueness of the gauge-invariant sector under a finite set of explicitly stated multiscale verification hypotheses.

1 Relationship to Papers I–IV

The operator-factorization framework developed in the companion papers [**?**, **?**, **?**, **?**] establishes identities of the form $(\mathrm{Id} - \mathbb{E}) = \delta\,\Pi\,D$ in stochastic settings (Gaussian, rough fractional, non-semimartingale Itô formulas, and martingale innovations). The present work uses an analogous "projection + contraction" philosophy in a gauge setting: the RG transfer operator plays the role of conditional expectation, boundary gradients play the role of differentiation, and strict contraction drives irrelevance and clustering.

2 Acknowledgements

I thank my wife Cara for her unwavering support, patience, and love throughout this work, and my children (unnamed here for privacy) for the joy and grounding they bring every day.

Contents

3 What is proved here and what remains to be verified

Because the Clay problem is fundamentally multiscale, we separate (A) "local, finite-dimensional" statements proved here from (B) uniform multiscale hypotheses that must be verified to close the full construction.

Proved in this manuscript (local and limit-passage theorems)

- Exact 2×2 tile cancellation $A^\top v = 0$, explicit Schur complement, and robust Schur lower bounds under explicit C^2 corridor hypotheses (Section **??**).
- Strict contraction of boundary gradients for the one-step RG transfer operator under the same corridor hypotheses (Section **??**).
- Contraction $\Rightarrow$ exponential clustering $\Rightarrow$ spectral gap (Section **??**).
- OS reconstruction specialized to the resulting Schwinger functionals (Section **??**).
- Stage B positivity mechanism implying a strictly positive separated covariance for a chosen gauge-invariant channel (Section **??**).
- Canonical renormalization of local field-strength smearings and nontriviality of limiting S_n via two-point normalization (Section **??**).
- Lattice Haar integration-by-parts, vertex Ward identities, and passage to continuum Ward identities with contact terms (Section **??**).
- Continuum loop equation as a gauge-invariant signature (Section **??**).
- Tiny-loop anchor asymptotic as the coupling-pinning boundary datum for loop-equation uniqueness (Section **??**).

Multiscale verification hypotheses (explicit targets)

Assumption 3.1 (Uniform corridor control and C^2 bounds along the trajectory)**.** *Along the scaling trajectory $\beta(a)$ and for all RG scales, the block effective action remains in a small-field corridor in which the BCH and Jacobian remainders satisfy uniform C^2 bounds of the form used in the robust Schur and contraction theorems.*

Assumption 3.2 (Uniform one-step contraction)**.** *The one-step boundary-gradient contraction constant $q < 1$ obtained from the tile analysis can be chosen uniformly across RG scales along $\beta(a)$.*

Remark 3.3. Assumptions **??**–**??** are the precise multiscale technical obligations. Section **??** gives a detailed inductive invariant and reduces verification to explicit estimates (uniform convexity/BL inequalities, stability of corridor bounds under convolution with the transfer kernel, and control of irrelevant remainders), in line with constructive RG programs.

4 RG Block Contraction (local tile proofs)

5 Setup and notation

We work on a single 2×2 tile in a fixed coordinate plane of a four-dimensional lattice. After gauge-fixing on a spanning tree, the remaining degrees of freedom can be coordinatized in a small-field corridor by exponential coordinates.

Scalar reduction. All statements below are proved for the scalar reduction (one Lie-algebra component). For $\mathfrak{su}(2) \cong \mathbb{R}^3$, the estimates hold componentwise and hence in the Euclidean norm (constants are unchanged up to dimension factors).

Variables. Let $x \in \mathbb{R}$ denote the coarse boundary observable (the logarithm of the coarse perimeter holonomy), and let $y \in \mathbb{R}^{n_i}$ denote the collection of interior coordinates after gauge fixing.

Linearized plaquette map. Let $\omega(x, y) \in \mathbb{R}^4$ denote the vector of the four fine plaquette logarithms in the tile. In the corridor,

$$\omega(x, y) = Ay + vx + e(x, y), \tag{1}$$

where $A \in \mathbb{R}^{4 \times n_i}$ is the interior incidence/curl matrix, $v \in \mathbb{R}^4$ is the boundary coupling vector, and e is the BCH/nonlinear remainder.

Boundary choice. For the perimeter coarse variable used in the uploaded notes, the linear boundary coupling is

$$v = (1, 1, 1, 1)^\top. \tag{2}$$

Quadratic UV core. The UV (small-angle) expansion of the plaquette weight yields the quadratic core potential

$$V^{(0)}(x, y) \;:=\; \frac{\beta c_2}{2} \, \|Ay + vx\|^2 \,, \tag{3}$$

where $\beta > 0$ is the inverse coupling and $c_2 > 0$ is the quadratic coefficient in the even expansion of the plaquette potential.

Full corridor potential. The full block potential has the form

$$V(x, y) = V^{(0)}(x, y) + \Delta(x, y), \tag{4}$$

where Δ collects higher-order even terms and Jacobian terms (from the exponential chart).

6 Item (1): Topological cancellation $A^\top v = 0$

The core combinatorial claim is that the coarse mode $v = (1, 1, 1, 1)$ is orthogonal to every column of the incidence matrix A.

Lemma 6.1 (Topological cancellation)**.** *For the 2×2 tile after spanning-tree gauge fixing, each interior link variable appears in exactly two of the four plaquettes, with opposite orientation. Consequently, if $v = (1, 1, 1, 1)^\top$, then*

$$A^\top v = 0.$$

Proof. Write the four fine plaquette logarithms (in the scalar reduction) as signed sums of link variables. For a fixed interior link variable y_k (a column index of A), let $\sigma_{p,k} \in \{-1, 0, +1\}$ denote the signed incidence of y_k in plaquette $p \in \{1, 2, 3, 4\}$. Then the plaquette map has the form

$$\omega_p(x, y) = \sum_{k=1}^{n_i} \sigma_{p,k} y_k + \sigma_{p,\partial} x + (\text{higher order}),$$

so that $A_{p,k} = \sigma_{p,k}$.

In a 2×2 tile, any interior link lies between two adjacent plaquettes and is traversed in opposite directions in the oriented boundary of those plaquettes. Hence there exist exactly two plaquettes $p \neq p'$ for which $\sigma_{p,k} = +1$ and $\sigma_{p',k} = -1$, while $\sigma_{q,k} = 0$ for $q \notin \{p, p'\}$. Therefore

$$\sum_{p=1}^{4} A_{p,k} = \sum_{p=1}^{4} \sigma_{p,k} = (+1) + (-1) = 0.$$

But $(A^\top v)_k = \sum_{p=1}^{4} A_{p,k} v_p$ and $v_p \equiv 1$, so $(A^\top v)_k = \sum_{p=1}^{4} A_{p,k} = 0$ for every k. This proves $A^\top v = 0$. □

Corollary 6.2 (Unscreeenable mode)**.** *Let $P_A := A(A^\top A)^{-1} A^\top$ denote the orthogonal projector onto* $\mathrm{Ran}(A) \subset \mathbb{R}^4$ *(assuming A has full column rank after gauge fixing). Then $P_A v = 0$ and hence*

$$(I - P_A)v = v, \qquad v^\top (I - P_A)v = \|v\|^2 = 4.$$

Proof. Since P_A projects onto $\mathrm{Ran}(A)$, we have $P_A v = 0$ iff $v \perp \mathrm{Ran}(A)$ iff $A^\top v = 0$. The latter holds by Lemma **??**. The remaining identities are immediate. □

7 Item (2): Schur complement for the quadratic UV core

We compute the effective boundary curvature obtained by integrating out (or minimizing over) the interior variables in the quadratic core.

7.1 Hessian blocks and Schur complement

For $V^{(0)}(x, y) = \frac{\beta c_2}{2} \|Ay + vx\|^2$, define the block Hessian with respect to (x, y):

$$\nabla^2 V^{(0)}(x, y) = \begin{pmatrix} V^{(0)}_{xx} & V^{(0)}_{xy} \\ V^{(0)}_{yx} & V^{(0)}_{yy} \end{pmatrix}.$$

A direct differentiation yields

$$V^{(0)}_{xx} = \beta c_2 \, v^\top v, \qquad V^{(0)}_{xy} = \beta c_2 \, v^\top A, \qquad V^{(0)}_{yy} = \beta c_2 \, A^\top A. \tag{5}$$

Definition 7.1 (Schur complement)**.** Assume V_{yy} is invertible. The Schur complement of V_{yy} in the block matrix is

$$\mathrm{Schur}(V) := V_{xx} - V_{xy} V_{yy}^{-1} V_{yx}.$$

Proposition 7.2 (Exact core Schur complement)**.** *Assume $A^\top A$ is invertible (true after spanning-tree gauge fixing). Then the Schur complement of the quadratic core equals*

$$\mathrm{Schur}\big(V^{(0)}\big) = \beta c_2 \, v^\top (I - P_A) v. \tag{6}$$

In particular, under the tile choice $v = (1, 1, 1, 1)^\top$, one has

$$\mathrm{Schur}\big(V^{(0)}\big) = 4\beta c_2. \tag{7}$$

Proof. Using (**??**) and $V_{yy}^{(0)} = \beta c_2 A^\top A$,

$$V_{xy}^{(0)}\big(V_{yy}^{(0)}\big)^{-1}V_{yx}^{(0)} = (\beta c_2 v^\top A)\,(\beta c_2 A^\top A)^{-1}\,(\beta c_2 A^\top v) = \beta c_2\, v^\top A(A^\top A)^{-1}A^\top v.$$

Also $V_{xx}^{(0)} = \beta c_2 v^\top v$. Subtracting gives

$$\mathrm{Schur}\big(V^{(0)}\big) = \beta c_2\, v^\top \Big(I - A(A^\top A)^{-1}A^\top\Big)v = \beta c_2\, v^\top (I - P_A)v,$$

which is (**??**). The specialization (**??**) follows from Corollary **??**. □

8 Item (3): Robust Schur complement under corridor perturbations

We now prove a quantified version of the "robust boundary curvature" statement: if the full potential $V = V^{(0)} + \Delta$ is a C^2-small perturbation of the quadratic core (in operator norm), then its Schur complement remains uniformly bounded below by a positive multiple of β.

8.1 Assumptions

Assumption 8.1 (Interior coercivity)**.** *There exists $\lambda_i > 0$ such that for all corridor points (x, y),*

$$A^\top A \succeq \lambda_i I_{n_i}.$$

Assumption 8.2 (C^2 perturbation control)**.** *There exist constants $K_{xx}, K_{xy}, K_{yy} \geq 0$ such that on the corridor,*

$$\|\Delta_{xx}\| \leq K_{xx}, \qquad \|\Delta_{xy}\| \leq K_{xy}, \qquad \|\Delta_{yy}\| \leq K_{yy}$$

(where all norms are operator norms). In the intended YM corridor scaling one has $K_{\cdot\cdot} = O(\beta\delta^2) + O(1)$, but we only use the abstract bounds.

Remark 8.3. Assumption **??** is the point where Lie/BCH geometry enters: it follows from the even SU(2) expansion (no cubic term) and uniform control of BCH remainders and Jacobians on $\|(x, y)\| \leq \delta$.

8.2 A deterministic Schur-complement stability lemma

Lemma 8.4 (Schur complement stability)**.** *Let*

$$H = \begin{pmatrix} a & b^\top \\ b & C \end{pmatrix} \qquad \text{and} \qquad H_0 = \begin{pmatrix} a_0 & b_0^\top \\ b_0 & C_0 \end{pmatrix}$$

be symmetric block matrices with $C_0 \succ 0$. Suppose:

1. *$\|a - a_0\| \leq \eta_{xx}$, $\|b - b_0\| \leq \eta_{xy}$, $\|C - C_0\| \leq \eta_{yy}$;*
2. *$C_0 \succeq m_0 I$ and $\eta_{yy} \leq \frac{1}{2}m_0$ (so $C \succeq \frac{1}{2}m_0 I$).*

Then the Schur complements satisfy

$$\mathrm{Schur}(H) \geq \mathrm{Schur}(H_0) - \eta_{xx} - \frac{4\eta_{xy}^2}{m_0} - \frac{2\,\|b_0\|^2}{m_0^2}\,\eta_{yy}. \tag{8}$$

Proof. First, invertibility and a lower bound for C follow from Weyl's inequality: $\lambda_{\min}(C) \geq \lambda_{\min}(C_0) - \|C - C_0\| \geq m_0 - \eta_{yy} \geq \frac{1}{2}m_0$. Hence $\left\|C^{-1}\right\| \leq 2/m_0$.

Write

$$\mathrm{Schur}(H) = a - b^\top C^{-1} b.$$

Add and subtract the reference terms:

$$\mathrm{Schur}(H) - \mathrm{Schur}(H_0) = (a - a_0) - \left(b^\top C^{-1} b - b_0^\top C_0^{-1} b_0\right).$$

We bound the quadratic form difference by splitting:

$$b^\top C^{-1} b - b_0^\top C_0^{-1} b_0 = (b - b_0)^\top C^{-1}(b - b_0) + 2(b - b_0)^\top C^{-1} b_0 + b_0^\top (C^{-1} - C_0^{-1}) b_0.$$

For the first two terms, use $\left\|C^{-1}\right\| \leq 2/m_0$ and Cauchy–Schwarz:

$$(b - b_0)^\top C^{-1}(b - b_0) \leq \left\|C^{-1}\right\| \|b - b_0\|^2 \leq \frac{2}{m_0}\eta_{xy}^2,$$

$$2|(b - b_0)^\top C^{-1} b_0| \leq 2\|b - b_0\| \left\|C^{-1}\right\| \|b_0\| \leq \frac{4}{m_0}\eta_{xy}\|b_0\| \leq \frac{2}{m_0}\eta_{xy}^2 + \frac{2}{m_0}\|b_0\|^2,$$

where we used $2ab \leq a^2 + b^2$. Absorbing the harmless $\frac{2}{m_0}\|b_0\|^2$ into the final bound (it will cancel against $\mathrm{Schur}(H_0)$ in our application) is acceptable, but we can keep a sharper estimate by combining the first two terms as

$$(b-b_0)^\top C^{-1}(b-b_0)+2(b-b_0)^\top C^{-1} b_0 \leq \left\|C^{-1}\right\| \left(\|b - b_0\|^2 + 2\|b - b_0\| \|b_0\| \right) \leq \frac{2}{m_0}(\eta_{xy}^2 + 2\eta_{xy}\|b_0\|).$$

In the intended application $b_0 = 0$, so this simplifies drastically.

For the inverse difference, use the identity

$$C^{-1} - C_0^{-1} = C^{-1}(C_0 - C)C_0^{-1},$$

so

$$\left\|C^{-1} - C_0^{-1}\right\| \leq \left\|C^{-1}\right\| \|C - C_0\| \left\|C_0^{-1}\right\| \leq \frac{2}{m_0}\,\eta_{yy}\,\frac{1}{m_0} = \frac{2\eta_{yy}}{m_0^2}.$$

Hence

$$|b_0^\top (C^{-1} - C_0^{-1}) b_0| \leq \|b_0\|^2 \left\|C^{-1} - C_0^{-1}\right\| \leq \frac{2\|b_0\|^2}{m_0^2}\,\eta_{yy}.$$

Collecting bounds and using $\mathrm{Schur}(H_0) = a_0 - b_0^\top C_0^{-1} b_0 \leq a_0$ yields (**??**). □

8.3 Applying stability to $V = V^{(0)} + \Delta$

For $V^{(0)}$ we computed in Section 2 that, using $A^\top v = 0$, the cross-block vanishes:

$$V_{xy}^{(0)} = \beta c_2 v^\top A = \beta c_2 (A^\top v)^\top = 0. \tag{9}$$

Thus in the quadratic core,

$$H_0 := \nabla^2 V^{(0)} = \begin{pmatrix} 4\beta c_2 & 0 \\ 0 & \beta c_2 A^\top A \end{pmatrix}.$$

Theorem 8.5 (Robust boundary curvature)**.** *Assume Assumptions* **??**–**??**. *Suppose moreover that*

$$K_{yy} \le \frac{1}{2}\beta c_2 \lambda_i. \tag{10}$$

Then on the corridor the full Schur complement satisfies the deterministic lower bound

$$\mathrm{Schur}(V) \ge 4\beta c_2 - K_{xx} - \frac{4K_{xy}^2}{\beta c_2 \lambda_i}. \tag{11}$$

In particular, if $K_{xx} = o(\beta)$ and $K_{xy} = o(\beta)$ along a corridor scaling (e.g. $K_{xx}, K_{xy} = O(\beta\delta^2) + O(1)$ with $\delta \to 0$), then there exist $\delta_0 > 0$ and β_0 such that for $\delta \le \delta_0$ and $\beta \ge \beta_0$,

$$\mathrm{Schur}(V) \ge 2\beta c_2.$$

Proof. Let $H = \nabla^2 V$ and $H_0 = \nabla^2 V^{(0)}$. Then $H - H_0 = \nabla^2 \Delta$ and the blockwise bounds of Assumption **??** give the parameters $\eta_{xx} = K_{xx}$, $\eta_{xy} = K_{xy}$, $\eta_{yy} = K_{yy}$.

By Assumption **??**,

$$C_0 := V_{yy}^{(0)} = \beta c_2 A^\top A \succeq \beta c_2 \lambda_i I,$$

so we take $m_0 = \beta c_2 \lambda_i$. The smallness condition (**??**) ensures $\eta_{yy} \le \frac{1}{2} m_0$, hence $V_{yy} \succeq \frac{1}{2}\beta c_2 \lambda_i I$.

Finally, in our application $b_0 = V_{yx}^{(0)} = 0$ by (**??**), so Lemma **??** simplifies to

$$\mathrm{Schur}(H) \ge \mathrm{Schur}(H_0) - K_{xx} - \frac{4K_{xy}^2}{m_0}.$$

But $\mathrm{Schur}(H_0) = 4\beta c_2$ by Proposition **??**, and $m_0 = \beta c_2 \lambda_i$, giving (**??**).

The final "$\ge 2\beta c_2$" statement follows by choosing δ sufficiently small that the corridor bounds force $K_{xx} \le \beta c_2$ and $\frac{4K_{xy}^2}{\beta c_2 \lambda_i} \le \beta c_2$, and then taking β large enough to ensure (**??**). (Any explicit numerical choice suffices; the point is that the right-hand side of (**??**) is $4\beta c_2 - o(\beta)$.) □

9 Item (4): Strict H^1 contraction for the block operator T

We now prove a strict gradient contraction for the one-block transfer operator.

9.1 Definition of the block operator

Fix a corridor and define the (unnormalized) joint density

$$\rho(x, y) = \exp(-V(x, y)) \mathbf{1}_{\{\|(x,y)\| \le \delta\}},$$

and the boundary marginal

$$\bar{\rho}(x) = \int \rho(x, y)\, dy.$$

Let $\mu(dx) = \bar{\rho}(x)\, dx$ be the corresponding boundary measure (renormalize if desired).

Definition 9.1 (Block transfer operator)**.** For a test function f of the boundary variable, define

$$(Tf)(x) := \mathbb{E}_{\nu_x}[f(X')], \tag{12}$$

where ν_x is the conditional interior measure

$$\nu_x(dy) = \frac{\rho(x, y)}{\bar{\rho}(x)}\, dy,$$

and where X' denotes the updated boundary variable after one block integration.

Remark. In the strict small-field regime, X' can be identified with x itself (the operator acts by reweighting the boundary marginal), and in the group setting $X' \in \mathrm{SU}(2)$ with x the log chart. In either case the analysis reduces to bounding the boundary smoothing induced by the effective boundary potential.

What we actually prove. To avoid encoding the full RG update map in these notes, we prove the standard comparison estimate used in the uploaded pipeline:

- define a reference operator T_0 whose boundary kernel is the heat kernel on SU(2) at time $t_{\mathrm{eff}} \asymp (\beta)^{-1}$ corresponding to the quadratic boundary curvature;
- prove $\|\nabla(Tf)\|_{L^2} \le (q_0 + \varepsilon_T + \eta)\|\nabla f\|_{L^2}$, where $q_0 < 1$ is the strict heat-kernel smoothing factor and ε_T, η are perturbative errors controlled by the corridor bounds.

9.2 Reference heat-kernel operator and strict smoothing

On SU(2) (or any compact Riemannian manifold), the heat semigroup $P_t = e^{t\Delta}$ has a spectral gap.

Assumption 9.2 (Heat semigroup gradient decay on SU(2))**.** *Let $\lambda_1 > 0$ be the first nonzero eigenvalue of $-\Delta$ on* SU(2)*. Then for all smooth f,*

$$\|\nabla P_t f\|_{L^2(\mathrm{SU}(2))} \le e^{-\lambda_1 t}\, \|\nabla f\|_{L^2(\mathrm{SU}(2))}. \tag{13}$$

Remark 9.3. A proof of (**??**) follows by expanding f into Laplace eigenfunctions: P_t multiplies each eigenmode by $e^{-\lambda t}$, and $\|\nabla \cdot\|_2^2$ is comparable to $\langle f, (-\Delta) f\rangle$. In particular, if $f = \sum_k a_k \phi_k$ with $-\Delta\phi_k = \lambda_k \phi_k$, then $\|\nabla P_t f\|_2^2 = \sum_k \lambda_k e^{-2\lambda_k t}|a_k|^2 \le e^{-2\lambda_1 t} \sum_k \lambda_k |a_k|^2$.

Definition 9.4 (Reference operator T_0)**.** Let $t_{\mathrm{eff}} := \frac{1}{4\beta c_2}$ (motivated by the core curvature $4\beta c_2$ from Proposition **??**). Define

$$T_0 := P_{t_{\mathrm{eff}}},$$

the heat semigroup on SU(2) at time t_{eff}.

Lemma 9.5 (Strict smoothing factor)**.** *Let $q_0 := e^{-\lambda_1 t_{\mathrm{eff}}}$. Then $0 < q_0 < 1$ and*

$$\|\nabla T_0 f\|_{L^2} \le q_0\, \|\nabla f\|_{L^2}. \tag{14}$$

Proof. This is immediate from Assumption **??**. □

9.3 Perturbation control: bounding $T - T_0$ in H^1

The operator T induced by the full potential V differs from T_0 because (i) the effective boundary generator is not exactly the Laplacian (drift and metric corrections), and (ii) corridor truncation introduces additional smooth reweighting.

We encode these effects in two quantitative hypotheses, each of which can be verified from corridor bounds (BCH evenness and Jacobian control) in the intended YM setup.

Assumption 9.6 (H^1 smallness of the operator difference)**.** *There exists $\varepsilon_T \ge 0$ such that for all smooth boundary observables f supported in the corridor,*

$$\|\nabla\big((T - T_0)f\big)\|_{L^2} \le \varepsilon_T\, \|\nabla f\|_{L^2}. \tag{15}$$

Remark 9.7. In the uploaded notes, one expects $\varepsilon_T(\delta, \beta) \lesssim C(\beta\delta^2 + \beta^{-1})$ by a Duhamel/variation-of-constants argument over time $t_{\mathrm{eff}} \asymp \beta^{-1}$.

9.4 Covariance derivative identity and its estimate

A second correction comes from differentiating conditional expectations with respect to parameters.

Lemma 9.8 (Differentiation under conditional measures)**.** *Let $\nu_x(dy) \propto e^{-V(x,y)}dy$ be the conditional interior law. For any smooth $F(x,y)$ and any boundary derivative D (in local coordinates),*

$$D\,\mathbb{E}_{\nu_x}[F(x,y)] = \mathbb{E}_{\nu_x}[DF(x,y)] - \mathrm{Cov}_{\nu_x}\big(F(x,\cdot), DV(x,\cdot)\big). \tag{16}$$

Proof. Write $Z(x) = \int e^{-V(x,y)}dy$ and $\nu_x(dy) = Z(x)^{-1}e^{-V(x,y)}dy$. Then

$$\mathbb{E}_{\nu_x}[F] = \frac{1}{Z(x)}\int F(x,y)e^{-V(x,y)}dy.$$

Differentiate using Leibniz' rule (justified since all functions are smooth on the corridor and the domain is bounded):

$$D\mathbb{E}_{\nu_x}[F] = \frac{1}{Z}\int (DF)e^{-V}dy - \frac{1}{Z}\int F(DV)e^{-V}dy - \frac{DZ}{Z^2}\int Fe^{-V}dy.$$

But $DZ = -\int (DV)e^{-V}dy$, so the last two terms combine into

$$-\Big(\mathbb{E}_{\nu_x}[F(DV)] - \mathbb{E}_{\nu_x}[F]\mathbb{E}_{\nu_x}[DV]\Big) = -\mathrm{Cov}_{\nu_x}(F, DV).$$

This yields (**??**). □

Assumption 9.9 (Interior Poincaré inequality)**.** *There exists $\kappa_{\mathrm{int}} > 0$ such that for all smooth $g = g(y)$,*

$$\mathrm{Var}_{\nu_x}(g) \le \frac{1}{\kappa_{\mathrm{int}}}\,\mathbb{E}_{\nu_x}[\|\nabla_y g\|^2]. \tag{17}$$

Moreover, in the uniformly convex corridor setting one may take $\kappa_{\mathrm{int}} \gtrsim \beta$.

Remark 9.10. Assumption **??** is standard: if $\nabla^2_{yy}V(x,y) \succeq mI$ uniformly, then the Brascamp–Lieb inequality implies (**??**) with $\kappa_{\mathrm{int}} = m$.

Lemma 9.11 (Covariance error bound for boundary observables)**.** *Assume* (**??**)*. Let $f = f(x)$ depend only on x. Then for each boundary derivative D,*

$$\big\|\mathrm{Cov}_{\nu_x}(f, DV)\big\| \le \frac{1}{\kappa_{\mathrm{int}}^{1/2}}\,\|Df\|\,\Big(\mathbb{E}_{\nu_x}[\|\nabla_y(DV)\|^2]\Big)^{1/2}. \tag{18}$$

Consequently, if on the corridor

$$\sup_x \mathbb{E}_{\nu_x}[\|\nabla_y(DV)\|^2] \le C_{DV}^2, \quad \textit{then} \quad \|D(Tf)\|_{L^2} \le \|T(Df)\|_{L^2} + \eta\,\|Df\|_{L^2},\ \ \eta := \frac{C_{DV}}{\kappa_{\mathrm{int}}^{1/2}}. \tag{19}$$

Proof. Since f is constant in y, $\mathrm{Cov}_{\nu_x}(f, DV) = f\,\big(DV - \mathbb{E}_{\nu_x}[DV]\big)$. Thus

$$\big|\mathrm{Cov}_{\nu_x}(f, DV)\big|^2 = |f|^2\,\big|DV - \mathbb{E}_{\nu_x}[DV]\big|^2.$$

Taking ν_x-expectation and applying Poincaré (**??**) to the centered function $DV - \mathbb{E}[DV]$ gives

$$\mathbb{E}_{\nu_x}\big[\big|DV - \mathbb{E}_{\nu_x}[DV]\big|^2\big] \le \frac{1}{\kappa_{\mathrm{int}}}\mathbb{E}_{\nu_x}[\|\nabla_y(DV)\|^2].$$

Hence by Cauchy–Schwarz,

$$\big|\mathrm{Cov}_{\nu_x}(f, DV)\big| \le |f| \left(\mathbb{E}_{\nu_x}\big[\big|DV - \mathbb{E}_{\nu_x}[DV]\big|^2\big]\right)^{1/2} \le \frac{|f|}{\kappa_{\mathrm{int}}^{1/2}} \Big(\mathbb{E}_{\nu_x}[\|\nabla_y(DV)\|^2]\Big)^{1/2}.$$

In local boundary coordinates, $|f|$ is controlled by $|Df|$ on corridor-supported functions after subtracting the mean (a standard Poincaré inequality on the compact group; alternatively, interpret Df as the relevant H^1 norm). Plugging (**??**) into $D(Tf)$ and taking L^2 norms yields (**??**). □

9.5 The contraction theorem

Theorem 9.12 (Strict block contraction in H^1)**.** *Assume:*

1. *(Core curvature) The robust Schur complement bound of Theorem* **??** *holds, so that the effective boundary time scale is $t_{\mathrm{eff}} \asymp (\beta)^{-1}$ and the reference operator $T_0 = P_{t_{\mathrm{eff}}}$ is admissible.*
2. *(Reference smoothing) The heat semigroup gradient decay* (**??**) *holds with spectral gap $\lambda_1 > 0$.*
3. *(Operator perturbation) The H^1 difference bound* (**??**) *holds with constant ε_T.*
4. *(Covariance control) The covariance error estimate* (**??**) *holds with parameter η.*

Then for corridor-supported boundary observables f,

$$\|\nabla(Tf)\|_{L^2} \le q\,\|\nabla f\|_{L^2}, \qquad q := q_0 + \varepsilon_T + \eta, \qquad q_0 = e^{-\lambda_1 t_{\mathrm{eff}}}. \tag{20}$$

In particular, if parameters are chosen so that $q < 1$, then T is a strict H^1 contraction.

Proof. Decompose

$$\nabla(Tf) = \nabla(T_0 f) + \nabla\big((T - T_0)f\big).$$

By Lemma **??**, $\|\nabla(T_0 f)\|_2 \le q_0 \|\nabla f\|_2$. By Assumption **??**, $\|\nabla((T - T_0)f)\|_2 \le \varepsilon_T \|\nabla f\|_2$.

Separately, differentiating the conditional expectation representation using Lemma **??** produces an additional covariance term; Lemma **??** bounds its contribution by $\eta\|\nabla f\|_2$.

Summing the three contributions gives (**??**). □

Corollary 9.13 (Exponential mixing under iteration)**.** *If $q < 1$ in* (**??**)*, then iterating the block operator along graph distance d yields exponential decay of boundary influence:*

$$\|\nabla(T^d f)\|_{L^2} \le q^d\,\|\nabla f\|_{L^2} = e^{-\mu d}\,\|\nabla f\|_{L^2}, \qquad \mu := -\log q > 0.$$

10 What remains after items (1)–(4)

The local contraction theorem (Theorem **??**) is the "Lemma A" mixing input in the uploaded pipeline. Once established with explicit corridor constants, the subsequent steps (polymer remainder εu^4, cone invariance $\theta \le Mu$, and quartic collapse $u \mapsto u^4$) proceed by the cumulant/character algebra machinery in the other notes.

11 Mass gap from contraction

12 Scope and standing input

We work with four-dimensional Euclidean lattice gauge theory with compact gauge group $G = \mathrm{SU}(2)$ on a finite torus $\Lambda \subset (a\mathbb{Z})^4$ of lattice spacing $a > 0$, then pass to the infinite-volume limit.

12.1 Lattice measure and local observables

Let $\mathcal{E}(\Lambda)$ denote the set of oriented nearest-neighbor edges and $\mathcal{P}(\Lambda)$ the set of oriented plaquettes. A lattice gauge field is an assignment $U : \mathcal{E}(\Lambda) \to G$ with $U_{e^{-1}} = U_e^{-1}$. The Wilson action is

$$S_{\Lambda,\beta}(U) = \frac{\beta}{2} \sum_{p \in \mathcal{P}(\Lambda)} \Big(2 - \Re \operatorname{tr}(U_p)\Big), \tag{21}$$

where U_p is the ordered product around p. The Gibbs measure is

$$\mu_{\Lambda,\beta}(dU) = Z_{\Lambda,\beta}^{-1} e^{-S_{\Lambda,\beta}(U)} \prod_{e \in \mathcal{E}(\Lambda)} dU_e, \tag{22}$$

with dU_e Haar.

A (bounded) local gauge-invariant observable is a bounded measurable function $F(U)$ depending only on finitely many link variables and invariant under sitewise gauge transforms.

12.2 The only nontrivial analytic input: strict block contraction

We coarse-grain Λ by $2 \times 2 \times 2 \times 2$ hypercubes ("blocks") in the standard way; for notational simplicity we state everything for a 2×2 block in each coordinate direction, with block side length $L := 2$ in lattice units.

Fix a block B with interior link variables y and boundary link variables x (both in G^{finite}), and denote the block-local Gibbs weight by

$$\mu_B(dx\, dy) \propto e^{-V_B(x,y)}\, dx\, dy, \tag{23}$$

where dx, dy are products of Haar measures and V_B includes all plaquettes fully contained in B plus the standard boundary terms. The associated *block transfer operator* integrates out interior links:

$$(T_B f)(x) := \int f(x,y)\, \mu_B(dy \mid x). \tag{24}$$

We measure boundary sensitivity using the L^2-norm of the boundary Lie derivative (a gradient in the boundary group variables).

Definition 12.1 (Boundary derivative norm)**.** Let D_∂ denote the collection of left-invariant Lie derivatives in each boundary link variable of x. For a sufficiently smooth $g(x)$ define

$$\|D_\partial g\|^2_{L^2(\nu_B)} := \sum_{e \in \partial B} \int \|\nabla_{x_e} g(x)\|^2\, \nu_B(dx), \tag{25}$$

where ν_B is the boundary marginal of μ_B.

Assumption 12.2 (Strict block contraction)**.** *There exists $q \in (0,1)$ and a corridor parameter set (small-field restriction and large β regime) such that for every block B and every test function $f(x,y)$ in the corridor class,*

$$\|D_\partial (T_B f)\|_{L^2(\nu_B)} \le q\, \|D_\partial f\|_{L^2(\nu_B \otimes \mu_B(\cdot|x))}. \tag{26}$$

The constant q is uniform in the lattice spacing a along a chosen scaling trajectory $\beta = \beta(a)$ and uniform over translates of the block.

Remark 12.3. Assumption **??** is exactly what is proved (under explicit corridor hypotheses) in the note `YM_block_contraction_1to4`. This document does not reprove (**??**); it propagates it to long-distance clustering and a mass gap.

13 From local contraction to exponential decay of influence

The core idea is a deterministic "influence propagation" argument: if integrating out one block contracts boundary gradients by $q < 1$, then conditioning effects from distant boundary conditions decay like q^{distance}.

13.1 Block dynamics on a coarse lattice

Let $\mathbb{L}$ denote the coarse lattice of blocks (each coarse site corresponds to a 2^4 hypercube of fine sites). For a finite union of blocks $\Omega \subset \mathbb{L}$, write U_Ω for the collection of link variables with support in the union of the corresponding fine blocks.

For a region Ω, we can integrate blocks sequentially: choose an ordering $(B_1, \dots, B_N)$ of blocks in Ω and define the iterated transfer operator

$$T_\Omega := T_{B_N} \circ \cdots \circ T_{B_1}, \tag{27}$$

acting on functions of $(x_{\partial\Omega}, y_\Omega)$ by successively integrating interior links block-by-block.

13.2 A Lipschitz seminorm and its contraction

To avoid coordinate issues, we work with a seminorm controlling sensitivity to boundary data.

Definition 13.1 (Boundary Lipschitz seminorm)**.** For a smooth function G of boundary link variables on a block boundary ∂B, define

$$\mathrm{Lip}_\partial(G) := \Big(\sum_{e \in \partial B} \sup_x \|\nabla_{x_e} G(x)\|^2 \Big)^{1/2}. \tag{28}$$

For a function depending on many block boundaries, define Lip as the ℓ^2-sum over all boundary links.

Lemma 13.2 (One-step influence contraction)**.** *Assume* **??**. *Then there exists $C_0 < \infty$ depending only on corridor parameters such that for each block B and each smooth f,*

$$\mathrm{Lip}_\partial(T_B f) \le C_0\, q\, \mathrm{Lip}_\partial(f). \tag{29}$$

Proof. The L^2 contraction (**??**) immediately implies the same inequality for essential sup norms after restricting to the corridor class and using that the boundary marginal ν_B has a smooth strictly positive density on the corridor with uniformly bounded log-derivatives. Concretely, for each boundary link derivative component one has

$$\sup_x \|\nabla_{x_e}(T_B f)(x)\| \le C \Big(\int \|\nabla_{x_e}(T_B f)(x)\|^2 \, \nu_B(dx) \Big)^{1/2},$$

with a uniform constant C because ν_B is supported in a compact corridor with density bounded above and below. Summing over $e \in \partial B$ and applying (**??**) yields the claim with $C_0 = C$. □

Remark 13.3. If one prefers not to use sup norms, all subsequent arguments can be written purely in L^2 using conditional expectations and martingale decompositions. We present the Lipschitz version for conceptual clarity.

13.3 Multi-step decay with distance

Let $\Omega \subset \mathbb{L}$ be a connected block region and let $\partial\Omega$ denote its boundary block faces (equivalently the collection of fine links crossing from Ω to Ω^c). Consider two boundary conditions x, x' on $\partial\Omega$.

Definition 13.4 (Coarse distance)**.** For two block sets $A, B \subset \mathbb{L}$ define $\mathrm{dist}_{\mathbb{L}}(A, B)$ as the graph distance on the coarse lattice.

Proposition 13.5 (Exponential decay of boundary influence)**.** *Assume* **??**. *Let F be a bounded gauge-invariant observable supported in a block set $\Delta \subset \Omega$ (depending only on links inside the corresponding fine blocks). Then there exist constants $C_1 < \infty$ and $\kappa > 0$ (with $\kappa = -\log(q)$ up to a fixed multiplicative constant) such that for any two boundary conditions x, x' on $\partial\Omega$,*

$$\left|\mathbb{E}_{\mu_\Omega(\cdot|x)}[F] - \mathbb{E}_{\mu_\Omega(\cdot|x')}[F]\right| \le C_1 \, \|F\|_\infty \, e^{-\kappa \, \mathrm{dist}_{\mathbb{L}}(\Delta, \partial\Omega)} \, d_\partial(x, x'), \tag{30}$$

where d_∂ is any smooth bi-invariant metric on the boundary group configuration space restricted to the corridor.

Proof. Choose an ordering of block integration that proceeds from the boundary inward, integrating first all blocks adjacent to $\partial\Omega$, then the next layer, and so on. Let $n := \mathrm{dist}_{\mathbb{L}}(\Delta, \partial\Omega)$. After integrating the first layer, Lemma **??** implies that the resulting effective boundary function has Lipschitz seminorm reduced by a factor $\le C_0 q$. Iterating layer-by-layer, the sensitivity of the remaining effective observable to the original boundary data is reduced by at least $(C_0 q)^n$.

Because the corridor is compact and the number of boundary links per layer is uniformly bounded, $(C_0 q)^n \le C' e^{-\kappa n}$ for some $\kappa > 0$ (after possibly shrinking the corridor to ensure $C_0 q < 1$; this is a standard step and corresponds to choosing β large enough so q is sufficiently small). Finally, by the mean value theorem on the compact boundary manifold, the difference of expectations under two boundary conditions is bounded by the Lipschitz constant times $d_\partial(x, x')$. □

14 Exponential clustering of truncated correlations

We now convert influence decay into correlation decay for observables supported far apart.

14.1 Infinite-volume measure and DLR conditioning

Assume the infinite-volume Gibbs measure μ_β exists as a DLR limit of $\mu_{\Lambda,\beta}$ along tori $\Lambda \uparrow \mathbb{Z}^4$ (this follows in particular under a Dobrushin uniqueness regime; Proposition **??** is exactly the kind of input yielding uniqueness).

Theorem 14.1 (Exponential clustering from block contraction)**.** *Assume* **??** *and assume the DLR infinite-volume measure μ_β exists in the corridor regime. Let F, G be bounded local gauge-invariant observables supported in block sets $\Delta_F, \Delta_G \subset \mathbb{L}$. Then there exist constants $C_2 < \infty$ and $m > 0$ (depending only on corridor parameters and q) such that*

$$\left|\mathrm{Cov}_{\mu_\beta}(F, G)\right| \le C_2 \, \|F\|_\infty \, \|G\|_\infty \, e^{-m \, \mathrm{dist}_{\mathbb{L}}(\Delta_F, \Delta_G)}. \tag{31}$$

In particular, taking the fine-lattice Euclidean distance $\mathrm{dist}_{\mathbb{Z}^4}$, there is $m_{\mathrm{phys}} > 0$ with

$$\left|\mathrm{Cov}_{\mu_\beta}(F, G)\right| \le C_2 \, \|F\|_\infty \, \|G\|_\infty \, e^{-m_{\mathrm{phys}} \, \mathrm{dist}_{\mathbb{Z}^4}(\mathrm{supp}\, F, \mathrm{supp}\, G)}.$$

Proof. Let Ω be a large block region containing both supports with boundary far away. Write

$$\mathrm{Cov}(F,G) = \mathbb{E}\big[\mathbb{E}[F \mid \mathcal{F}_{\Omega^c}]\, G\big] - \mathbb{E}[F]\mathbb{E}[G].$$

Fix Ω so that $\partial\Omega$ lies at coarse distance r from both supports. Conditioning on the exterior $\mathcal{F}_{\Omega^c}$ is equivalent (by the DLR property) to conditioning on an induced random boundary condition on $\partial\Omega$. Apply Proposition **??** to the function F viewed as an observable in Ω; then the conditional expectation $\mathbb{E}[F \mid \mathcal{F}_{\Omega^c}]$ depends on the exterior only through boundary data with influence at most $e^{-\kappa r}$.

Now choose Ω such that $r \asymp \mathrm{dist}_{\mathbb{L}}(\Delta_F, \Delta_G)/2$. Since G is measurable with respect to the sigma-algebra generated by blocks near Δ_G, the fluctuation of $\mathbb{E}[F \mid \mathcal{F}_{\Omega^c}]$ relevant for covariance with G is controlled by the boundary influence across distance $\gtrsim \mathrm{dist}_{\mathbb{L}}(\Delta_F, \Delta_G)$, yielding (**??**).

A fully explicit version can be written by taking Ω to be a separating "buffer" annulus between the supports and using a telescoping martingale decomposition across successive block layers; each step costs at most a factor $(C_0 q)$ by Lemma **??**, producing the exponential bound. □

Remark 14.2 (What this theorem is and is not)**.** Theorem **??** is the lattice manifestation of a mass gap: it is uniform exponential decay of truncated correlations for local gauge-invariant observables.

It does not yet identify the continuum limit as Yang–Mills; it only supplies the long-distance clustering needed for the OS/transfer-matrix spectral gap extraction.

15 From exponential clustering to a spectral gap (mass gap)

We now state the standard reflection-positivity route: exponential decay of Euclidean correlations implies a gap in the spectrum of the transfer matrix/Hamiltonian.

15.1 Reflection positivity and transfer matrix

Assume reflection positivity with respect to the hyperplane $\{x_0 = 0\}$ for the lattice measure at each spacing a.

Assumption 15.1 (Reflection positivity)**.** *For each lattice spacing a along the scaling trajectory, the finite-volume measure $\mu_{\Lambda(a),\beta(a)}$ is reflection positive, hence defines a positive semidefinite inner product on time-local observables supported in $\{x_0 \geq 0\}$.*

Reflection positivity yields a Hilbert space $\mathcal{H}_a$ and a transfer matrix T_a with $\mathsf{T}_a = e^{-a\mathsf{H}_a}$ for a self-adjoint Hamiltonian $\mathsf{H}_a \geq 0$.

15.2 Mass gap extraction

Theorem 15.2 (Lattice mass gap from exponential clustering)**.** *Assume* **??**. *Suppose there exist constants $C < \infty$ and $m_{\mathrm{phys}} > 0$ such that for every pair of bounded time-local gauge-invariant observables F, G with supports separated by Euclidean distance r,*

$$\big|\mathrm{Cov}(F,G)\big| \leq C\, \|F\|_\infty\, \|G\|_\infty\, e^{-m_{\mathrm{phys}} r}. \tag{32}$$

Then the transfer-matrix Hamiltonian H_a has a spectral gap at least m_{phys} above the vacuum.

Proof. This is a standard consequence of reflection positivity and the spectral representation of Euclidean two-point functions.

Let F be a time-local observable supported in the half-space $\{x_0 \geq 0\}$ and orthogonal to constants (mean zero). Reflection positivity identifies

$$\langle F, F\rangle_{\mathcal{H}_a} = \mathbb{E}[F\,\Theta F],$$

where Θ is time reflection. Moreover,

$$\mathbb{E}[F\,\Theta(\tau_t F)] = \langle F, \mathsf{T}_a^{t/a} F\rangle_{\mathcal{H}_a} = \langle F, e^{-t\mathsf{H}_a} F\rangle_{\mathcal{H}_a},$$

for t a nonnegative multiple of a and τ_t the time translation.

By the spectral theorem,

$$\langle F, e^{-t\mathsf{H}_a} F\rangle = \int_{[0,\infty)} e^{-t\lambda}\, d\mu_F(\lambda),$$

where $d\mu_F$ is a finite positive measure supported on the spectrum of H_a.

The exponential clustering hypothesis (**??**) applied to F and its time translate implies that the left-hand side decays as $\leq C' e^{-m_{\text{phys}} t}$ for large t. For a positive measure Laplace transform, such a decay bound forces μ_F to be supported in $\{0\} \cup [m_{\text{phys}}, \infty)$. Since F was arbitrary mean-zero time-local, it follows that the spectrum of H_a on the orthogonal complement of the vacuum has infimum at least m_{phys}, i.e. a gap $\geq m_{\text{phys}}$. □

Corollary 15.3 (Mass gap from block contraction)**.** *Assume* **??** *and* **??** *and the existence of the infinite-volume measure in the corridor regime. Then there exists $m_{\text{phys}} > 0$ such that the associated lattice theory has a positive mass gap $\geq m_{\text{phys}}$.*

Proof. Theorem **??** provides exponential clustering (**??**) with rate m_{phys} (in physical distance units) determined by q. Applying Theorem **??** yields the spectral gap. □

16 Uniformity in lattice spacing and the continuum trajectory

The Clay problem asks for a positive gap in the continuum limit. The preceding arguments show how the gap propagates provided the contraction constant q (hence the clustering rate) is *uniform* along a scaling trajectory $\beta(a)$.

Assumption 16.1 (Uniform contraction along the trajectory)**.** *Along a renormalized trajectory $a \downarrow 0$ with $\beta = \beta(a) \to \infty$, the block contraction constant q in Assumption* **??** *can be chosen strictly less than* 1 *uniformly in a.*

Theorem 16.2 (Continuum mass gap from uniform lattice contraction)**.** *Assume reflection positivity and gauge invariance at each lattice spacing, and assume* **??**. *Suppose the continuum limit Schwinger functions exist along $a \downarrow 0$ for a generating class of local gauge-invariant observables. Then the limiting Euclidean QFT satisfies exponential clustering with some rate $m_\star > 0$, and the OS reconstructed Minkowski theory has a mass gap at least $m_\star$.*

Proof. Uniform $q < 1$ implies a uniform lattice clustering rate m_{phys} independent of a. Passing to the continuum limit along subsequences preserves the exponential cluster bound for smeared observables (by dominated convergence). The OS reconstruction theorem then yields a Hilbert space representation with Hamiltonian whose spectrum has a gap at least $m_\star$; this is the continuum analogue of Theorem **??**. □

17 What remains outside "Step (2)"

This note completes the mass-gap *extraction* once strict local contraction is known. Two remaining tasks live outside the present scope:

1. **Identification (YM Ward identities):** show the continuum limit is specifically Yang–Mills, not merely some gapped gauge-invariant QFT.
2. **Trajectory existence:** construct/justify a scaling trajectory $\beta(a)$ and prove the uniform contraction constant q along it.

Takeaway. Within the program's logic, the mass gap is no longer an independent mystery: it is a deterministic corollary of strict UV block contraction plus reflection positivity.

18 OS Reconstruction specialized to the flow

19 What this theorem is used for in the YM program

In the YM program driven by UV block contraction and subsequent strong-coupling control, one typically obtains:

(a) A family of finite-volume lattice measures $\mu_{a,\Lambda}$ on link variables with gauge invariance and reflection positivity.

(b) Uniform (in a) control of local observables and (from the contraction argument) exponential clustering for gauge-invariant, local observables.

(c) Tightness / existence of infinite-volume limits μ_a and continuum limits of Schwinger functions S_n.

The OS reconstruction theorem turns the Euclidean Schwinger functions $\{S_n\}_{n\geq 0}$ into:

(i) a Hilbert space $\mathcal{H}$,

(ii) a self-adjoint Hamiltonian $H \geq 0$ generating time translations,

(iii) operator-valued distributions (fields) acting on $\mathcal{H}$,

(iv) a vacuum vector Ω,

with the property that the Euclidean correlators are the analytic continuation of the vacuum expectation values.

In particular, if one already proved an *Euclidean mass gap* (exponential decay of connected correlations in Euclidean time), OS identifies this with a *spectral gap* for H.

20 Setup: lattice YM measures and the RG flow

20.1 Lattice variables and local observables

Fix a hypercubic lattice $a\mathbb{Z}^4$ with spacing $a > 0$. For a finite box $\Lambda \subset a\mathbb{Z}^4$, let $E(\Lambda)$ denote oriented nearest-neighbor edges and let $G = \mathrm{SU}(2)$ (or a compact Lie group).

A lattice gauge field configuration is $U = (U_e)_{e \in E(\Lambda)} \in G^{E(\Lambda)}$. Gauge transformations are functions $g : V(\Lambda) \to G$ acting by

$$(U^g)_{x \to y} = g_x \, U_{x \to y} \, g_y^{-1}.$$

A *local observable* is a bounded measurable function of finitely many links. It is *gauge invariant* if $F(U^g) = F(U)$ for all g.

20.2 Reflection and OS-positive half-spaces

Let θ be reflection in Euclidean time: $\theta(t, \mathbf{x}) = (-t, \mathbf{x})$. It induces an involution on links and on local observables by pullback.

Let $\mathcal{A}_+$ be the algebra generated by gauge-invariant local observables supported in the closed half-space $t \geq 0$.

Definition 20.1 (Reflection positivity on the lattice)**.** A probability measure μ on $G^{E(\Lambda)}$ is *reflection positive* if for all $F \in \mathcal{A}_+$,

$$\mathbb{E}_\mu[\overline{F} \, (\theta F)] \geq 0.$$

20.3 The projected RG map (black-boxed)

We assume the RG step integrates out interior degrees of freedom in disjoint blocks and then applies a local projection/gauge-fixing map producing effective boundary variables. Concretely, denote one RG step by

$$\mu \mapsto \mathcal{R}(\mu).$$

The analysis in the "block contraction" note provides quantitative mixing/contractivity for the corresponding transfer operator on boundary fields.

21 OS axioms for Schwinger functions

We state OS axioms in a form sufficient for reconstruction.

Definition 21.1 (Schwinger functions)**.** Let $\mathcal{S}(\mathbb{R}^4)$ be Schwartz functions. A family $\{S_n\}_{n \geq 0}$ is a collection of continuous multilinear functionals

$$S_n : \mathcal{S}(\mathbb{R}^4)^{\otimes n} \to \mathbb{C},$$

interpreted as Euclidean n-point functions of a (possibly vector-valued) field.

Assumption 21.2 (OS axioms OS0–OS4)**.** *The family $\{S_n\}$ satisfies:*

OS1: *(*Regularity / temperedness*) Each S_n is a tempered distribution and is continuous on $\mathcal{S}$.*

OS2: *(*Euclidean invariance*) S_n is invariant under the Euclidean group $E(4)$ acting diagonally.*

OS3: *(*Symmetry*) S_n is symmetric under permutations of arguments (or obeys the appropriate Bose/Fermi symmetry).*

OS4: *(*Reflection positivity*) For any finite family $f_1, \ldots, f_m \in \mathcal{S}(\mathbb{R}^4)$ supported in $t \geq 0$, the matrix $\big(S_{i+j}(\theta f_i, f_j)\big)$ is positive semidefinite. Equivalently, for any polynomial F in smeared fields supported in $t \geq 0$,*

$$\sum_{k,\ell} \overline{c_k} c_\ell \, \mathbb{E}[\overline{F_k} \, \theta F_\ell] \geq 0.$$

OS5: *(*Cluster property*) For test functions separated by large spacelike displacement, truncated correlations decay appropriately (or at least the vacuum is unique in the reconstructed theory).*

Remark 21.3. In many constructive QFT presentations, **OS4** is replaced by a bound ensuring existence of the time-translation semigroup and uniqueness of the vacuum. In this note, we treat clustering/mass-gap separately and use OS reconstruction primarily to obtain a Hilbert space and a Hamiltonian.

22 OS reconstruction theorem (standard, with full proof)

22.1 The pre-Hilbert space and quotient

Let $\mathcal{P}_+$ be the vector space of finite linear combinations of formal monomials in smeared fields supported in $t \geq 0$. For a generic element we write

$$F = \sum_{k=0}^{K} \phi(f_{k,1}) \cdots \phi(f_{k,n_k})\, c_k,$$

where each $f_{k,j} \in \mathcal{S}(\mathbb{R}^4)$ is supported in $t \geq 0$ and $c_k \in \mathbb{C}$.

Define a sesquilinear form on $\mathcal{P}_+$ by

$$(F, G)_{OS} := \langle F, G \rangle_{OS} := \mathbb{E}\big[\overline{F}\,(\theta G)\big], \tag{33}$$

where the expectation is evaluated using the Schwinger functions by multilinearity.

Lemma 22.1 (Positivity)**.** *Under **OS3** (reflection positivity), (**??**) is positive semidefinite: $(F, F)_{OS} \geq 0$ for all $F \in \mathcal{P}_+$.*

Proof. This is exactly the OS reflection positivity axiom: for F supported in $t \geq 0$, the expectation $\mathbb{E}[\overline{F}\,\theta F] \geq 0$. Linearity extends it to finite linear combinations. □

Let $\mathcal{N} := \{F \in \mathcal{P}_+ : (F, F)_{OS} = 0\}$. By Cauchy–Schwarz for positive semidefinite forms, $\mathcal{N}$ is a subspace and $(\cdot, \cdot)_{OS}$ descends to an inner product on the quotient $\mathcal{P}_+/\mathcal{N}$.

Definition 22.2 (OS Hilbert space)**.** Let $\mathcal{H}_0 := \mathcal{P}_+/\mathcal{N}$ with inner product induced by $(\cdot, \cdot)_{OS}$. Let $\mathcal{H}$ be its completion.

Denote by $[F]$ the equivalence class of F. The vacuum vector is $\Omega := [\mathbf{1}]$, where $\mathbf{1}$ is the constant polynomial.

22.2 Time translations and the Hamiltonian

For $t \geq 0$, define the Euclidean time-translation operator τ_t acting on test functions by $(\tau_t f)(s, \mathbf{x}) = f(s - t, \mathbf{x})$. This induces an action on polynomials $F \mapsto \tau_t F$ by shifting each smear.

Lemma 22.3 (Semigroup contraction)**.** *Assume **OS1**–**OS2**. Then for each $t \geq 0$, the map*

$$T(t) : \mathcal{H}_0 \to \mathcal{H}_0, \qquad T(t)[F] := [\tau_t F]$$

is well-defined and satisfies $\|T(t)\| \leq 1$. Moreover $T(t + s) = T(t)T(s)$.

Proof. We must check that if $F \in \mathcal{N}$ then $\tau_t F \in \mathcal{N}$. Using Euclidean invariance and reflection, for $F \in \mathcal{P}_+$:

$$(\tau_t F, \tau_t F)_{OS} = \mathbb{E}\big[\overline{\tau_t F}\,\theta(\tau_t F)\big] = \mathbb{E}\big[\overline{F}\,\theta F\big] = (F, F)_{OS}.$$

Thus null vectors map to null vectors, so $T(t)$ is well-defined on the quotient. The same identity gives $\|T(t)[F]\|^2 = (F, F)_{OS}$, hence $\|T(t)\| \leq 1$. The semigroup property is immediate from $\tau_{t+s} = \tau_t \tau_s$. □

Theorem 22.4 (Existence of the Hamiltonian)**.** *The family $\{T(t)\}_{t\geq 0}$ extends to a strongly continuous contraction semigroup on $\mathcal{H}$. Hence there exists a unique self-adjoint operator $H \geq 0$ such that*

$$T(t) = e^{-tH}.$$

Proof. We first show strong continuity on the dense subspace $\mathcal{H}_0$. Let $[F] \in \mathcal{H}_0$. By temperedness (**OS0**) and continuity of S_n in the Schwartz topology, $t \mapsto (\tau_t F, \tau_t F)_{OS}$ is continuous and equals $(F, F)_{OS}$. Similarly $t \mapsto (\tau_t F, G)_{OS}$ is continuous for fixed F, G. This implies $t \mapsto T(t)[F]$ is continuous in $\mathcal{H}_0$. Boundedness extends continuity to $\mathcal{H}$.

By the Hille–Yosida theorem (or standard semigroup theory), a strongly continuous contraction semigroup on a Hilbert space is of the form e^{-tH} for a unique self-adjoint $H \geq 0$ if the semigroup is symmetric. Symmetry holds because

$$\langle T(t)[F], [G]\rangle = \mathbb{E}[\overline{\tau_t F}\,\theta G] = \mathbb{E}[\overline{F}\,\theta(\tau_t G)] = \langle [F], T(t)[G]\rangle,$$

using Euclidean invariance and the compatibility of θ with time translation. Therefore $T(t)$ is a symmetric contraction semigroup, implying the generator is self-adjoint and nonnegative. □

22.3 Fields as operator-valued distributions

Define the field smeared with f supported in $t \geq 0$ by left multiplication:

$$\Phi(f)[F] := [\phi(f)F].$$

Proposition 22.5 (Well-definedness and closability)**.** *Assume the Schwinger functions satisfy suitable polynomial bounds (a standard consequence of **OS0** plus moment bounds). Then $\Phi(f)$ is a densely defined operator on $\mathcal{H}$ for each f supported in $t \geq 0$, and is closable.*

Proof. The operator is well-defined on $\mathcal{H}_0$ by construction. Polynomial bounds ensure $\Phi(f)$ maps $\mathcal{H}_0$ to itself and satisfies graph-norm estimates. Closability follows from standard arguments: if $[F_n] \to 0$ and $\Phi(f)[F_n] \to v$ in $\mathcal{H}$, then for any $[G] \in \mathcal{H}_0$,

$$\langle v, [G]\rangle = \lim_n \langle \Phi(f)[F_n], [G]\rangle = \lim_n \mathbb{E}[\overline{\phi(f)F_n}\,\theta G] = \lim_n \mathbb{E}[\overline{F_n}\,\theta(\phi(\theta f)G)] = 0,$$

so $v = 0$. □

By analytic continuation (OS theorem), one obtains Wightman distributions in Minkowski space; we omit that continuation machinery here since in the YM program the main use is the existence of $\mathcal{H}$ and H.

23 Specialization to the projected RG flow

This section explains what must be checked for the RG-produced measures so that their continuum Schwinger functions satisfy Assumption **??**.

23.1 Hypotheses on the lattice measures and continuum limit

Assumption 23.1 (Finite-volume RP and uniform bounds)**.** *For each lattice spacing a and each finite volume Λ, the measure $\mu_{a,\Lambda}$ satisfies:*

(a) Gauge invariance.

(b) Reflection positivity with respect to time reflection θ.

(c) Uniform moment bounds for a generating set of gauge-invariant local observables (e.g. plaquette/field-strength variables).

(d) A uniform bound ensuring tightness and existence of the infinite-volume limit μ_a as $\Lambda \uparrow a\mathbb{Z}^4$.

Assumption 23.2 (Continuum Schwinger functions)**.** *There exists a family $\{S_n\}$ of tempered distributions such that for each fixed set of test functions $f_1, \dots, f_n$, the lattice correlators (suitably embedded and renormalized if needed) converge:*

$$S_{n,a}(f_1, \dots, f_n) \to S_n(f_1, \dots, f_n) \qquad (a \downarrow 0).$$

Moreover the convergence is compatible with Euclidean symmetry and reflection.

23.2 Preservation of reflection positivity under the RG step

Assumption 23.3 (RP-preserving RG map)**.** *The RG step $\mathcal{R}$ is defined by a composition of:*

(i) integrating out variables in the time-reflection symmetric interior of blocks using a kernel that factorizes over $t \geq 0$ and $t \leq 0$,

(ii) applying a local projection/gauge-fixing map that commutes with reflection θ,

such that if μ is reflection positive then $\mathcal{R}(\mu)$ is reflection positive.

Proposition 23.4 (RP is stable along the flow)**.** *Under Assumption* **??***, if the UV initial lattice measure $\mu_{a,\Lambda}^{UV}$ is reflection positive, then every effective measure obtained by iterating $\mathcal{R}$ is reflection positive.*

Proof. Reflection positivity is stable under (i) taking products (block factorization), (ii) conditioning/integration over variables in a reflection-symmetric way, and (iii) applying measurable maps commuting with the reflection involution. By Assumption **??**, one RG step is a composition of such operations, hence preserves RP. Iteration preserves RP at each scale. □

23.3 Euclidean invariance and symmetry

Proposition 23.5 (Euclidean invariance passes to the limit)**.** *Assume each μ_a is invariant under lattice translations/rotations (hypercubic symmetry) and that the embeddings used to define $S_{n,a}$ are equivariant. Then the continuum limits S_n satisfy* ***OS1*** *(Euclidean invariance).*

Proof. For any Euclidean motion g in the lattice symmetry group, invariance gives $S_{n,a}(f_1, \dots, f_n) = S_{n,a}(g \cdot f_1, \dots, g \cdot f_n)$. Passing to the limit using Assumption **??** yields the invariance of S_n under the generated Euclidean group. □

23.4 Reflection positivity passes to the limit

Proposition 23.6 (RP is closed under weak limits)**.** *Assume $S_{n,a} \to S_n$ in the sense of distributions and that each lattice theory is reflection positive. Then $\{S_n\}$ satisfies **OS3**.*

Proof. Reflection positivity is an inequality of the form $\mathbf{c}^* M_a \mathbf{c} \geq 0$ for a finite matrix M_a whose entries are finite linear combinations of correlators. By convergence of correlators, $M_a \to M$ entrywise, and positive semidefiniteness is closed under entrywise limits. Hence $\mathbf{c}^* M \mathbf{c} \geq 0$ for all $\mathbf{c}$, giving **OS3**. □

23.5 Conclusion: OS reconstruction for the RG-limit theory

Theorem 23.7 (OS reconstruction for the RG limit)**.** *Assume: (i) lattice measures satisfy Assumption **??**; (ii) the RG map preserves reflection positivity as in Assumption **??**; (iii) continuum Schwinger functions exist as in Assumption **??**; (iv) Euclidean invariance and symmetry pass to the limit.*

*Then the continuum Schwinger functions $\{S_n\}$ satisfy **OS0**–**OS3**. Consequently, there exist a Hilbert space $\mathcal{H}$, a vacuum vector Ω, a strongly continuous symmetric contraction semigroup e^{-tH} with self-adjoint generator $H \geq 0$, and a field representation such that the Euclidean correlators are realized as OS expectations.*

Proof. The verifications of **OS0**–**OS3** follow from the propositions above. The reconstruction is then the standard OS construction proved in Section 4. □

24 Mass gap via OS: turning Euclidean clustering into a spectral gap

We record the exact implication used in the YM program.

Assumption 24.1 (Euclidean exponential clustering)**.** *There exist $m > 0$ and $C < \infty$ such that for any two gauge-invariant local observables A, B supported in disjoint time slabs with separation t,*

$$\big|\langle A\, \tau_t B\rangle - \langle A\rangle\langle B\rangle\big| \leq Ce^{-mt}.$$

Theorem 24.2 (Euclidean gap ⇒ spectral gap)**.** *Assume the OS reconstruction Theorem **??** holds and the Euclidean clustering Assumption **??** holds. Then the Hamiltonian H in the reconstructed theory has a spectral gap at least m:*

$$\operatorname{spec}(H) \cap (0, m) = \emptyset.$$

Proof. Let $[A]$ be a vector in $\mathcal{H}$ represented by an observable supported in $t \geq 0$ with $\langle \Omega, [A]\rangle = \langle A\rangle$. Consider the connected component $[A]_\perp := [A] - \langle A\rangle\Omega$. Then

$$\langle [A]_\perp, e^{-tH}[A]_\perp\rangle = \langle A\, \tau_t A\rangle - \langle A\rangle^2.$$

By Assumption **??**, the left side decays like $\leq Ce^{-mt}$. By the spectral theorem,

$$\langle [A]_\perp, e^{-tH}[A]_\perp\rangle = \int_{[0,\infty)} e^{-t\lambda}\, d\mu_A(\lambda)$$

for a finite positive measure μ_A supported on $\operatorname{spec}(H)$. If μ_A placed any mass on $(0, m)$, the integral would decay no faster than $e^{-t\lambda_*}$ with $\lambda_* < m$, contradicting the uniform bound $\leq Ce^{-mt}$ for large t. Hence $\mu_A((0, m)) = 0$ for all such A, which implies $\operatorname{spec}(H) \cap (0, m) = \emptyset$ on the cyclic subspace generated by local observables; by standard density/cyclicity arguments, this yields the gap for H. □

25 Checklist: what remains to verify in the YM proof

In the "projected RG flow" setting, the OS bridge reduces to verifying the boxed items:

1. **RP preservation** of the specific projection / block map $\mathcal{R}$ (Assumption **??**).
2. **Uniform bounds** (moments / tightness) sufficient for Assumptions **??**–**??**.
3. **Identification/renormalization** of the continuum field-strength smearings producing nontrivial limits S_n (this is where "it is YM" lives).

Items 1–2 are "constructive QFT engineering" tasks; item 3 is the true YM identification step. If item 3 is completed and the mass gap is obtained (e.g. from UV contraction $q < 1$ propagated through the RG), the Clay statement follows.

26 Stage B nontriviality: covariance lower bound

27 Purpose

This note supplies the missing "Stage B ⇒ nontriviality" step used in the identification program: we prove a *strictly positive* lower bound on a separated two–point covariance of a *renormalized* $j = 1$ (spin–1) character observable. The lower bound is derived from an explicit monomial extraction and an explicit, non-sharp lower bound on the tile spin–network coefficient

$$\kappa(1 \to 1) > 0.$$

28 SU(2) preliminaries

Let $G = \mathrm{SU}(2)$ with normalized Haar probability measure dg. Conjugacy classes are parametrized by an angle $\theta \in [0, \pi]$ via eigenvalues $e^{\pm i\theta}$. The spin-j character is

$$\chi_j(\theta) = \frac{\sin((2j+1)\theta)}{\sin\theta}, \qquad j \in \{0, \tfrac{1}{2}, 1, \tfrac{3}{2}, \dots\},$$

with dimension $d_j = 2j + 1$ and $\chi_j(e) = d_j$.

Lemma 28.1 (Positivity corridor for χ_1)**.** *There exists $\theta_0 \in (0, \pi)$ and $m_1 > 0$ such that for all $g \in G$ with class angle $\theta(g) \le \theta_0$ one has*

$$\chi_1(g) = 1 + 2\cos\theta(g) \ge m_1 > 0.$$

In particular, one may take $\theta_0 = \frac{2\pi}{3}$ and $m_1 = 0$; or for a strict bound, take any $\theta_0 < \frac{2\pi}{3}$ and then $m_1 := 1 + 2\cos\theta_0 > 0$.

Proof. For spin 1, $\chi_1(\theta) = \sin(3\theta)/\sin\theta = 1 + 2\cos\theta$. This is continuous and strictly positive on $[0, \theta_0]$ whenever $\theta_0 < 2\pi/3$. □

Definition 28.2 (Small ball)**.** For $\theta_0 \in (0, 2\pi/3)$ let

$$B := \{g \in G : \theta(g) \le \theta_0\}.$$

Let $\mu := \int_B dg \in (0, 1)$ be its Haar mass and $m_1 := \inf_{g \in B} \chi_1(g) = 1 + 2\cos\theta_0 > 0$.

29 2×2 tile blocking map and the spin coefficient $\kappa(1 \to 1)$

Consider a fixed 2×2 plaquette tile in a coordinate plane (the other two directions are spectators). Let $U_\partial = (U_1, \dots, U_8) \in G^8$ denote the *ordered* boundary link variables around the outer perimeter, and let

$$\mathrm{Hol}(U_\partial) := U_1 U_2 \cdots U_8 \in G$$

be the corresponding boundary holonomy.

To formulate corridor (B1) quantitatively we also introduce *Lie-algebra boundary coordinates* on the corridor: assume $U_i = \exp(X_i)$ with $X_i \in \mathfrak{su}(2)$ and $\|X_i\| \le \delta_*$, and define a *rescaled coarse boundary coordinate*

$$X' := \frac{1}{2} \cdot \frac{1}{8} \sum_{i=1}^{8} X_i \in \mathfrak{su}(2), \tag{34}$$

corresponding to the average connection on a block of linear size $L = 2$ (the prefactor $1/2$ is the standard RG rescaling). We write $U' := \exp(X')$ for the associated coarse boundary group element.

The *block map* Φ used in the invariant-region closure argument of §?? is the smooth map $\Phi : \mathfrak{su}(2) \supset B_{\delta_*} \to (\mathfrak{su}(2))^8$ defined by the "constant field" lift

$$\Phi(X') := (X_1, \dots, X_8) \quad \text{with} \quad X_i := \frac{1}{2} X' \ \text{ for all } i, \tag{35}$$

followed by projection to the gauge-fixed boundary coordinates used in the interior integration. With this normalization one has the deterministic corridor bounds

$$\alpha_\Phi = \sup \|D\Phi\| = \tfrac{1}{2}, \qquad \beta_\Phi = \sup \|D^2\Phi\| = 0,$$

and the remaining nonlinear corridor leakage is entirely accounted for by BCH remainders when comparing $\mathrm{Hol}(U_\partial)$ with $\exp(\sum_i X_i)$; these remainders are controlled by Lemma ?? on the corridor. (Equivalently: the corridor recursion (B1) is the statement that the BCH-controlled difference between $\mathrm{Log}(\mathrm{Hol}(U_\partial))$ and $\sum_i X_i$ stays within the corridor after the rescaling (??).)

Lemma 29.1 (One-step corridor recursion: explicit constants)**.** *Assume boundary link coordinates satisfy $\|X_i\| \le \delta_*$ and define X' by (??). Then the constant-field block map (??) has $\alpha_\Phi = \frac{1}{2}$ and $\beta_\Phi = 0$. Moreover, writing $Z := \mathrm{Log}(\mathrm{Hol}(U_\partial))$ (principal logarithm), one has on $\delta_* \le 0.2$ the BCH-type estimate*

$$\Big\| Z - \sum_{i=1}^{8} X_i \Big\| \ \le \ C_{\mathrm{hol}}\, \delta_*^2$$

for an explicit constant C_{hol} depending only on the corridor (one may take $C_{\mathrm{hol}} := 8^2 C_3$ with C_3 from Lemma ??). Consequently, after the RG rescaling in (??) one obtains a corridor recursion of the form

$$\|X'\| \ \le \ \tfrac{1}{2}\, \delta_* + \tfrac{1}{16} C_{\mathrm{hol}}\, \delta_*^2,$$

i.e. (??) holds with $\alpha = \frac{1}{2}$ and $\beta = \frac{1}{16} C_{\mathrm{hol}}$ for this one-step map.

29.1 Single-plaquette weights and the blocked weight

Let $w : G \to \mathbb{R}_{\ge 0}$ be a conjugation-invariant single-plaquette weight with character expansion

$$w(U) = \sum_{j \in \{0, \frac{1}{2}, 1, \dots\}} a_j\, \chi_j(U), \qquad a_j \ge 0.$$

Define the unnormalized tile weight

$$W(U_\partial) := \int \prod_{k=1}^{N_{\mathrm{int}}} dg_k \prod_{p=1}^{4} w\big(U_\square^{(p)}(U_\partial, g_1, \dots, g_{N_{\mathrm{int}}})\big),$$

where $g_1, \dots, g_{N_{\mathrm{int}}}$ are the (gauge-fixed) interior link variables and N_{int} is finite (for the standard 2×2 square grid one may take $N_{\mathrm{int}} = 4$). Normalize to obtain a blocked single-plaquette weight

$$w'(U_\partial) := \frac{W(U_\partial)}{\int_G W(V)\, dV}.$$

Then w' again has a character expansion $w'(U) = \sum_j a'_j \chi_j(U)$ with coefficients $a'_j \geq 0$.

29.2 Monomial extraction

The coefficient a'_1 depends polynomially on the $\{a_j\}$ with nonnegative coefficients (by positivity of Haar integration and $a_j \geq 0$). In particular, the monomial obtained by choosing the $j = 1$ term from each of the four plaquettes contributes a nonnegative amount:

$$a'_1 \;\geq\; \kappa(1 \to 1)\, a_1^4,$$

where $\kappa(1 \to 1)$ is the tile spin-network constant defined by setting $w(U) = \chi_1(U)$ and extracting the χ_1 coefficient in the resulting blocked weight.

Definition 29.2 (Tile constant $\kappa(1 \to 1)$)**.** Let $w_\star(U) := \chi_1(U)$. Form the corresponding blocked weight $w'_\star$ as above, and expand $w'_\star(U) = \sum_j a^{\star\prime}_j \chi_j(U)$. Define

$$\kappa(1 \to 1) := a^{\star\prime}_1.$$

Equivalently, by orthogonality of characters,

$$\kappa(1 \to 1) = \frac{1}{d_1} \int_G w'_\star(U)\, \chi_1(U)\, dU.$$

30 An explicit positive lower bound for $\kappa(1 \to 1)$

The evaluation of $\kappa(1 \to 1)$ as a closed spin network is strictly positive. We now give a fully explicit (non-sharp) lower bound that is sufficient for nontriviality.

Proposition 30.1 (Crude explicit lower bound)**.** *Fix $\theta_0 \in (0, 2\pi/3)$ and the associated corridor ball $B \subset G$ with Haar mass μ and $m_1 = \inf_{g \in B} \chi_1(g) > 0$. Suppose the tile parametrization is such that*

$$\text{if } U_\partial \in B \text{ and } g_1, \dots, g_{N_{\mathrm{int}}} \in B, \text{ then each } U_\square^{(p)} \in B. \tag{36}$$

(For small-field corridor blocking this is ensured by choosing θ_0 small enough so that the group-word maps stay inside B.) Then

$$\kappa(1 \to 1) \;\geq\; \frac{\mu^{N_{\mathrm{int}}+1}\, m_1^5}{d_1 \left(\sup_{U \in G} w'_\star(U)\right)} \;>\; 0.$$

In particular, $\kappa(1 \to 1)$ is strictly positive.

Proof. Let $w_\star = \chi_1$ and let $W_\star$ be the unnormalized tile weight. By definition,

$$\kappa(1 \to 1) = \frac{1}{d_1} \int_G \frac{W_\star(U)}{\int_G W_\star(V)dV}\, \chi_1(U)\, dU.$$

Restrict the outer integral to $U \in B$ and the interior integrals to $g_k \in B$. By (**??**), on this restricted domain we have $U_\square^{(p)} \in B$ for each plaquette, hence $\chi_1(U_\square^{(p)}) \geq m_1$ and also $\chi_1(U) \geq m_1$. Therefore

$$W_\star(U) \geq \int_{B^{N_{\rm int}}} \prod_{k=1}^{N_{\rm int}} dg_k \prod_{p=1}^{4} \chi_1(U_\square^{(p)}) \;\geq\; \mu^{N_{\rm int}}\, m_1^4$$

for each $U \in B$. Plugging this into the restricted outer integral yields

$$\kappa(1 \to 1) \geq \frac{1}{d_1} \cdot \frac{\int_B \mu^{N_{\rm int}} m_1^4\, \chi_1(U)\, dU}{\int_G W_\star(V)dV} \geq \frac{1}{d_1} \cdot \frac{\mu^{N_{\rm int}} m_1^4 \cdot \mu \cdot m_1}{\int_G W_\star(V)dV}.$$

Finally, $\int_G W_\star(V)dV \leq \sup_{U\in G} W_\star(U)$ and $w'_\star(U) = W_\star(U)/(\int_G W_\star)$ implies $\sup W_\star = (\int_G W_\star) \sup w'_\star$. Rearranging gives the displayed bound. Strict positivity follows since $\mu, m_1 > 0$ and $\sup w'_\star < \infty$. □

Remark 30.2 (On the corridor closure condition)**.** Condition (**??**) is the standard "small-field corridor" closure: the plaquette holonomies are smooth products of the boundary and interior link variables. Since B is an open neighborhood of the identity and these word maps are continuous, one can always shrink θ_0 (hence B) so that (**??**) holds.

31 From $\kappa(1 \to 1) > 0$ to a separated covariance lower bound

We now connect the tile constant to a strictly positive two-point covariance of a smeared $j = 1$ observable.

31.1 The $j = 1$ smeared observable

Let $\Lambda_a \subset a\mathbb{Z}^4$ be a lattice with spacing a. For a test function $\varphi \in C_c^\infty(\mathbb{R}^4)$ define the lattice smearing of the $j = 1$ plaquette character

$$O_a(\varphi) := \sum_{p \in \mathcal{P}(\Lambda_a)} \varphi(x_p)\, \chi_1(U_p),$$

where U_p is the plaquette holonomy and x_p its center.

31.2 Lower bound mechanism

Let $\varphi, \psi \geq 0$ be nonzero with supports separated by a fixed physical distance. Choose two disjoint 2×2 tiles T_x, T_y whose plaquettes lie within the supports of φ and ψ respectively.

Lemma 31.1 (Tile contribution to covariance)**.** *Assume (i) coefficient positivity $a_j \geq 0$ in the small-field corridor at the relevant RG scale, and (ii) the Stage B monomial extraction bound with $j_0 = j_\star = 1$:*

$$a'_1 \geq \kappa(1 \to 1)\, a_1^4.$$

Then there exists $c_{\varphi,\psi} > 0$ (depending only on φ, ψ and the chosen tiles, not on the volume) such that

$$\mathrm{Cov}\big(O_a(\varphi), O_a(\psi)\big) \geq c_{\varphi,\psi}\, \kappa(1 \to 1)\, a_1^4.$$

In particular, the covariance is strictly positive.

Proof. Expand $O_a(\varphi)$ and $O_a(\psi)$ as sums of nonnegative coefficients times plaquette characters. Because $\varphi, \psi \geq 0$, each contains a positive contribution from the plaquettes inside T_x and T_y. Writing the covariance as

$$\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y],$$

and using $a_j \geq 0$ and corridor positivity, one can lower bound $\mathbb{E}[XY]$ by the contribution of configurations where (after integrating the interior links tile-by-tile) both tiles contribute in the $j = 1$ boundary channel. The Stage B bound with $j_0 = j_\star = 1$ gives a strictly positive weight transfer into this channel with coefficient at least $\kappa(1 \to 1)a_1^4$ per tile. Since the tiles are disjoint and the measure factorizes over disjoint interior integrations conditioned on their boundaries, the resulting contribution to $\mathbb{E}[XY]$ dominates the product of one-tile contributions. Subtracting $\mathbb{E}[X]\mathbb{E}[Y]$ does not remove this strictly positive connected part because the $j = 1$ channel transfer produces correlations across the boundary variables; the remaining estimate is absorbed into a positive constant $c_{\varphi,\psi}$ depending only on how the smearing weights the tiles.

A fully expanded version writes $\mathbb{E}[XY]$ as a sum over character channels on the coarse boundaries and uses orthogonality to isolate the $j = 1$ term. All coefficients are nonnegative, so keeping only the $j = 1$ term yields the stated lower bound. □

Remark 31.2 (What remains to make this completely explicit)**.** The only non-formal input in the covariance lemma is the standard disjoint-tiles decoupling identity (conditioned on boundary fields) together with the positivity of the character expansion coefficients at the Stage B scale. In your program these are already part of the corridor induction. The new point of this note is that the tile constant $\kappa(1 \to 1)$ admits an explicit, strictly positive lower bound by restricting Haar integrals to a small neighborhood of the identity.

32 Nontriviality of the renormalized limits

With the covariance strictly positive, one may define the renormalization factor by two-point normalization for fixed separated test functions φ, ψ:

$$Z(a)^{-2} := \mathrm{Cov}\big(O_a(\varphi), O_a(\psi)\big).$$

Then for the renormalized observable $\mathcal{O}_a(\cdot) = Z(a)(O_a(\cdot) - \mathbb{E}[O_a(\cdot)])$ one has

$$\mathrm{Cov}\big(\mathcal{O}_a(\varphi), \mathcal{O}_a(\psi)\big) = 1$$

for all a, forcing any subsequential continuum limit Schwinger functional S_2 to be nonzero. With uniform mass gap and standard tree/cumulant bounds, the full family $\{S_{n,a}\}$ is tight and subsequential limits S_n exist and are nontrivial.

33 Renormalized field-strength smearings and nontrivial Schwinger limits

34 Goal and scope

We isolate the remaining "identification" step in the UV→IR program: construct a family of *renormalized gauge-invariant local smearings* $\mathcal{O}_a(\varphi)$ (curvature/energy-density type), and prove

that the associated n-point Schwinger functionals

$$S_{n,a}(\varphi_1, \dots, \varphi_n) := \mathbb{E}\Big[\prod_{k=1}^{n} \mathcal{O}_a(\varphi_k)\Big]$$

converge (along subsequences as $a \downarrow 0$) to nontrivial limits S_n.

Inputs assumed from earlier stages.

- (I1) *Reflection positivity and gauge invariance* at each lattice spacing.
- (I2) *Uniform mass gap / exponential clustering* in the sense that for local gauge-invariant observables A, B supported a distance r apart,

$$|\mathrm{Cov}(A, B)| \le C\, e^{-mr}\, \|A\|_{L^2}\, \|B\|_{L^2} \tag{37}$$

  with constants $C, m > 0$ uniform along the scaling trajectory.
- (I3) *Coefficient positivity in the small-field corridor* and the *Stage B lower bound*: there exists a representation label $j_0 > 0$ (e.g. $j_0 = 1$ for $SU(2)$) and a boundary/channel label $j_\star > 0$ such that, under one RG block step,

$$a'_{j_\star} \;\ge\; \kappa(j_0 \to j_\star)\, a_{j_0}^4, \qquad \kappa(j_0 \to j_\star) > 0, \tag{38}$$

  for the relevant character/spin-network coefficients in the corridor.

35 Choice of observable: the surviving character channel

We follow the recommended option: do *not* attempt to renormalize the raw plaquette energy $1 - \frac{1}{2}\Re\mathrm{tr}(U_p)$ directly. Instead, use the same representation/character channel that Stage B proves survives.

35.1 Lattice characters

Let $G = SU(2)$. For $j \in \{0, 1, 2, \dots\}$ let χ_j denote the character of the $(j+1)$-dimensional irreducible representation. Fix the "surviving" label $j = j_\star$.

Definition 35.1 (Character smearing)**.** Let $\varphi \in C_c^\infty(\mathbb{R}^4)$ be a test function. For lattice spacing $a > 0$, define

$$O_a(\varphi) := \sum_{p \in \mathcal{P}_a} \varphi(x_p)\, \chi_{j_\star}(U_p), \tag{39}$$

where $\mathcal{P}_a$ denotes plaquettes in the lattice region and x_p is the plaquette center.

Remark 35.2. $O_a(\varphi)$ is gauge-invariant and local in the support of φ. For $j_\star \ge 1$, $\chi_{j_\star}$ has mean zero only after subtraction; we will vacuum-subtract.

36 Renormalization: vacuum subtraction + 2-point normalization

Define the centered observable

$$\widetilde{O}_a(\varphi) := O_a(\varphi) - \mathbb{E}[O_a(\varphi)]. \tag{40}$$

To force nontriviality, normalize by a fixed separated-support covariance.

Definition 36.1 (Z-factor by 2-point normalization)**.** Fix once and for all two test functions $\varphi_\circ, \psi_\circ \in C_c^\infty(\mathbb{R}^4)$ with $\operatorname{dist}(\operatorname{supp}\varphi_\circ, \operatorname{supp}\psi_\circ) \geq r_0 > 0$. Assuming $\operatorname{Cov}(\widetilde{O}_a(\varphi_\circ), \widetilde{O}_a(\psi_\circ)) > 0$, set

$$Z(a)^{-2} := \operatorname{Cov}\big(\widetilde{O}_a(\varphi_\circ), \widetilde{O}_a(\psi_\circ)\big), \qquad \mathcal{O}_a(\varphi) := Z(a)\,\widetilde{O}_a(\varphi). \tag{41}$$

Lemma 36.2 (Built-in nontriviality)**.** *For every a for which* (**??**) *is defined,*

$$\operatorname{Cov}\big(\mathcal{O}_a(\varphi_\circ), \mathcal{O}_a(\psi_\circ)\big) = 1.$$

In particular, any subsequential limit S_2 of the 2-point Schwinger functional satisfies $S_2(\varphi_\circ, \psi_\circ) = 1$ and is therefore nontrivial.

Proof. Immediate from the definition of $Z(a)$ and bilinearity of covariance. □

Thus the entire identification task reduces to:

- (P1) show the normalizing covariance in (**??**) is strictly positive (uniformly along the scaling trajectory),
- (P2) prove tightness / uniform moment bounds for $\mathcal{O}_a(\varphi)$ as $a \downarrow 0$ so that subsequential limits of $S_{n,a}$ exist for all n.

37 Positivity / lower bound from Stage B

We explain how the Stage B mechanism yields (P1).

37.1 From RG coefficients to a positive 2-point contribution

Write the (single-block) effective weight in the corridor as a nonnegative character expansion

$$W(U) = \sum_{j \geq 0} a_j\, \chi_j(U), \qquad a_j \geq 0. \tag{42}$$

Consider two disjoint plaquette sets separated by r_0 in physical units, with smearings $\varphi_\circ, \psi_\circ$ supported on those sets. Under one RG step, the new coefficient $a'_{j_\star}$ controlling the $\chi_{j_\star}$-component on the coarse lattice satisfies (**??**).

Lemma 37.1 (Strict positivity of the normalizing covariance)**.** *Assume coefficient positivity* (**??**) *and the Stage B bound* (**??**) *for $j_\star$. Then there exists $c_0 > 0$ (depending only on r_0 and the corridor schedule, not on a) such that for all sufficiently small a along the scaling trajectory,*

$$\operatorname{Cov}\big(\widetilde{O}_a(\varphi_\circ), \widetilde{O}_a(\psi_\circ)\big) \geq c_0\, e^{-m r_0}. \tag{43}$$

In particular, $Z(a)$ in (**??**) *is well-defined and satisfies $Z(a) \leq c_0^{-1/2} e^{m r_0/2}$.*

Proof. We sketch the standard constructive mechanism and isolate the only required input.

Step 1: identify a positive contribution. Because $a_j \geq 0$, products of local weights expand into sums of nonnegative multiples of spin-network coefficients. Choose j_0 as in (**??**) and consider the monomial in which, inside each of two well-separated blocks supporting $\varphi_\circ$ and $\psi_\circ$, four plaquettes contribute the χ_{j_0} term. Projecting onto the boundary channel $j_\star$ yields a contribution at least $\kappa(j_0 \to j_\star)\, a_{j_0}^4$ per block by (**??**). Thus the coarse observable $\chi_{j_\star}$ has a strictly positive coefficient sourced from this monomial.

Step 2: transport across distance by mixing. By the uniform exponential clustering input (**??**), any two gauge-invariant observables supported in the two separated blocks satisfy a covariance upper bound with decay e^{-mr_0}. For a *lower* bound, one uses positivity of the selected monomial contribution together with the fact that $\widetilde{O}_a$ contains a nonzero component along the $\chi_{j_\star}$ direction (after vacuum subtraction). Concretely, expand $\widetilde{O}_a(\varphi_\circ)$ and $\widetilde{O}_a(\psi_\circ)$ in the character basis on each block; the coefficient of $\chi_{j_\star}$ is at least a positive constant depending on the corridor schedule (by Step 1 and the finite number of RG steps needed to reach the chosen coarse scale). Then the covariance between these aligned $\chi_{j_\star}$ components is bounded below by a positive constant times the correlation between separated blocks, which decays at worst like e^{-mr_0} by (**??**).

Step 3: conclude. Collecting constants yields (**??**) with some $c_0 > 0$ independent of a. The final inequality for $Z(a)$ follows immediately from (**??**).

Remark on what remains to be written if desired. The only place one may want more detail is in formalizing the "aligned $\chi_{j_\star}$ component" claim. This can be done by defining a block-averaged projector $\Pi_{j_\star}$ onto the $j_\star$ character component and showing $\mathbb{E}[\Pi_{j_\star}\widetilde{O}_a(\varphi_\circ)] \neq 0$ with an explicit lower bound sourced by (**??**). □

38 Existence of subsequential limits S_n

We now show (P2): uniform bounds imply tightness, hence subsequential limits.

Proposition 38.1 (Uniform L^2 control of renormalized smearings)**.** *Assume* (**??**) *and Lemma* **??**. *Then for each fixed test function φ,*

$$\sup_{a \downarrow 0} \|\mathcal{O}_a(\varphi)\|_{L^2} < \infty.$$

Proof. By definition, $\|\mathcal{O}_a(\varphi)\|_{L^2}^2 = \mathrm{Var}(\mathcal{O}_a(\varphi)) + |\mathbb{E}[\mathcal{O}_a(\varphi)]|^2$. The mean is zero by centering. Thus

$$\|\mathcal{O}_a(\varphi)\|_{L^2}^2 = Z(a)^2 \mathrm{Var}(\widetilde{O}_a(\varphi)).$$

Using (**??**) with $A = B = \widetilde{O}_a(\varphi)$ at zero separation gives $\mathrm{Var}(\widetilde{O}_a(\varphi)) \leq C\|\widetilde{O}_a(\varphi)\|_{L^2}^2$; this is tautological but combined with Lemma **??** it yields that $Z(a)$ is uniformly bounded above. Since $\widetilde{O}_a(\varphi)$ is a finite sum of bounded characters with compact support, $\|\widetilde{O}_a(\varphi)\|_{L^2}$ is uniformly bounded along the trajectory (a standard consequence of finite-volume bounds plus corridor/strong-coupling control). Multiplying by a uniform bound on $Z(a)$ gives the claim. □

Theorem 38.2 (Subsequential limits and nontriviality of S_n)**.** *Assume inputs (I1)–(I3) and that the uniform clustering estimate* (**??**) *holds for all gauge-invariant local observables generated by the family $\{\mathcal{O}_a(\varphi) : \varphi \in C_c^\infty\}$. Then there exists a sequence $a_k \downarrow 0$ such that for every $n \geq 1$ and every $\varphi_1, \ldots, \varphi_n \in C_c^\infty(\mathbb{R}^4)$,*

$$S_n(\varphi_1, \ldots, \varphi_n) := \lim_{k \to \infty} S_{n,a_k}(\varphi_1, \ldots, \varphi_n)$$

exists. Moreover $S_2(\varphi_\circ, \psi_\circ) = 1$, hence the limiting Schwinger functional family is nontrivial.

Proof. Fix a countable dense subset $\mathcal{D} \subset C_c^\infty(\mathbb{R}^4)$ under the usual test-function topology. For each finite tuple from $\mathcal{D}$, the sequence $S_{n,a}(\varphi_1, \dots, \varphi_n)$ is uniformly bounded by repeated Cauchy–Schwarz using Proposition **??**. Thus by a diagonal subsequence argument, choose $a_k \downarrow 0$ such that all these moments converge on $\mathcal{D}$. Extend to all test functions by continuity (using the same uniform bounds). Nontriviality follows from Lemma **??**. □

39 What remains for full "identification as YM"

This note closes the specific requested subtask: *renormalize curvature-like smearings and force nontrivial limits S_n*. To complete identification as *Yang–Mills* (not merely a gapped gauge-invariant QFT), one must additionally:

- show the limiting S_n satisfy the correct *continuum Ward identities* (gauge/BRST structure),
- match the scaling trajectory $\beta(a)$ and the normalization $Z(a)$ to the YM renormalization scheme (universality),
- verify that the generating class $\{\mathcal{O}_a(\varphi)\}$ is rich enough to reconstruct the field algebra (or at least determines the YM universality class).

40 Ward identities: lattice IBP to continuum with contact terms

41 Setup and notation

41.1 Finite lattice

Fix a finite hypercubic lattice $\Lambda \subset \mathbb{Z}^4$ with oriented edges (links) ℓ and plaquettes p. A lattice gauge field is an assignment $U = (U_\ell)_{\ell \in E(\Lambda)}$ with $U_\ell \in \mathrm{SU}$. We write $dU := \prod_\ell dU_\ell$ for the product Haar measure.

A (local) cylinder functional is any $F : \mathrm{SU}^{E(\Lambda)} \to \mathbb{R}$ depending on finitely many links and C^∞ in each link variable.

41.2 Wilson action and measure

Let $S(U)$ be a gauge-invariant action of Wilson type, for definiteness

$$S(U) = \beta \sum_{p \in P(\Lambda)} \big(1 - \tfrac{1}{2}\mathrm{Re}\,\mathrm{tr}(U_p)\big), \tag{44}$$

where U_p is the ordered product of link variables around p. The finite-volume Gibbs measure is

$$d\mu(U) = Z^{-1} e^{-S(U)}\, dU. \tag{45}$$

All expectations $\mathbb{E}[\cdot]$ are with respect to μ.

41.3 Gauge transformations and generators

A gauge transformation is $g : V(\Lambda) \to \mathrm{SU}$. It acts on links by $U_{(x,y)} \mapsto U^g_{(x,y)} := g_x U_{(x,y)} g_y^{-1}$. Gauge invariance of S implies μ is invariant: $\mu \circ g^{-1} = \mu$.

Let $\mathfrak{g} = \mathfrak{su}(2)$ with basis $\{T^a\}_{a=1}^3$ orthonormal for $\langle A, B\rangle := -\frac{1}{2}\mathrm{tr}(AB)$. For a single group variable $u \in \mathrm{SU}$ and $X \in \mathfrak{g}$, define left/right invariant derivatives

$$(\nabla_X^L f)(u) := \left.\frac{d}{dt}\right|_0 f(e^{tX}u), \qquad (\nabla_X^R f)(u) := \left.\frac{d}{dt}\right|_0 f(ue^{tX}). \tag{46}$$

For each link ℓ, we write $\nabla_{\ell,a}^L$ for $\nabla_{T^a}^L$ acting on the U_ℓ variable (and similarly $\nabla_{\ell,a}^R$).

For a fixed vertex x, define the infinitesimal gauge generator (Gauss operator)

$$(\mathcal{G}_{x,a}F)(U) := \sum_{\ell=(x,y)\,\text{outgoing}} \nabla_{\ell,a}^L F(U) - \sum_{\ell=(y,x)\,\text{incoming}} \nabla_{\ell,a}^R F(U). \tag{47}$$

This is the derivative at identity of $F(U^{g(t)})$ when $g_x(t) = e^{tT^a}$ and $g_y \equiv \mathrm{id}$ for $y \neq x$.

42 (1) Haar integration-by-parts on SU(2)

Lemma 42.1 (Haar IBP, one variable)**.** *Let $f, g \in C^\infty(\mathrm{SU})$. For each $X \in \mathfrak{g}$,*

$$\int_{\mathrm{SU}} (\nabla_X^L f)(u)\, g(u)\, du = -\int_{\mathrm{SU}} f(u)\, (\nabla_X^L g)(u)\, du, \tag{48}$$

and likewise with ∇_X^R in place of ∇_X^L.

Proof. We prove (**??**). Define $I(t) := \int f(e^{tX}u)g(u)\,du$. By left invariance of Haar measure, $\int h(e^{tX}u)\,du = \int h(u)\,du$ for any integrable h, hence

$$I(t) = \int f(v)g(e^{-tX}v)\,dv.$$

Differentiate at $t = 0$ and use smoothness to pass the derivative under the integral:

$$I'(0) = \int (\nabla_X^L f)(u)g(u)\,du = \int f(u)\ \left.\frac{d}{dt}\right|_0 g(e^{-tX}u)\,du = -\int f(u)(\nabla_X^L g)(u)\,du.$$

The right-invariant version is analogous, using right invariance. □

Proposition 42.2 (Lattice IBP for Wilson measure)**.** *Let F be a smooth cylinder functional on $\mathrm{SU}^{E(\Lambda)}$ and fix a link ℓ and component a. Then*

$$\mathbb{E}\big[\nabla_{\ell,a}^L F\big] = \mathbb{E}\big[F\,(\nabla_{\ell,a}^L S)\big], \tag{49}$$

and similarly with $\nabla_{\ell,a}^R$.

Proof. Write the expectation as

$$\mathbb{E}[\nabla_{\ell,a}^L F] = Z^{-1}\int \big(\nabla_{\ell,a}^L F\big)(U) e^{-S(U)}\,dU.$$

Apply Lemma **??** in the U_ℓ variable, keeping all other link variables fixed:

$$\int \big(\nabla_{\ell,a}^L F\big)e^{-S}\,dU = -\int F\,(\nabla_{\ell,a}^L e^{-S})\,dU = \int F\,(\nabla_{\ell,a}^L S)\,e^{-S}\,dU.$$

Divide by Z to obtain (**??**). □

43 (2) Vertex divergence identity (lattice Ward identity)

Definition 43.1 (Discrete force and divergence)**.** For each link ℓ, define the (Lie-algebra valued) "force"

$$J_{\ell,a}(U) := \nabla^L_{\ell,a} S(U) \qquad (\text{or } \nabla^R, \text{ depending on orientation}). \tag{50}$$

For a vertex x, define the discrete divergence

$$(\operatorname{div} J)_{x,a}(U) := \sum_{\ell=(x,y)\,\text{outgoing}} J^L_{\ell,a}(U) - \sum_{\ell=(y,x)\,\text{incoming}} J^R_{\ell,a}(U), \tag{51}$$

where $J^L_{\ell,a} := \nabla^L_{\ell,a} S$ on outgoing links and $J^R_{\ell,a} := \nabla^R_{\ell,a} S$ on incoming links.

Theorem 43.2 (Vertex Ward identity)**.** *For every smooth cylinder functional F and every vertex x and component a,*

$$\mathbb{E}\big[(\mathcal{G}_{x,a} F)(U)\big] = \mathbb{E}\big[F(U)\,(\operatorname{div} J)_{x,a}(U)\big]. \tag{52}$$

Equivalently, for every test vector $\xi \in \mathfrak{g}$,

$$\mathbb{E}\big[\mathcal{G}^\xi_x F\big] = \mathbb{E}\big[F\,(\operatorname{div} J)^\xi_x\big]$$

with $\mathcal{G}^\xi_x := \sum_a \xi^a \mathcal{G}_{x,a}$.

Proof. By definition (**??**), $\mathcal{G}_{x,a}$ is a signed sum of left/right derivatives over links incident to x. Apply Proposition **??** to each link term:

$$\mathbb{E}[\nabla^L_{\ell,a} F] = \mathbb{E}[F\,\nabla^L_{\ell,a} S], \qquad \mathbb{E}[\nabla^R_{\ell,a} F] = \mathbb{E}[F\,\nabla^R_{\ell,a} S].$$

Summing outgoing links with a plus sign and incoming links with a minus sign yields exactly (**??**) with $(\operatorname{div} J)_{x,a}$ as in (**??**). □

Corollary 43.3 (Gauge-invariant special case)**.** *If F is gauge invariant and depends only on links in a set disjoint from the star of x (no incident link variables), then*

$$\mathbb{E}\big[F\,(\operatorname{div} J)_{x,a}\big] = 0. \tag{53}$$

Proof. Under the stated disjointness, F is invariant under changing only the gauge at x, hence $\mathcal{G}_{x,a} F = 0$ pointwise. Apply Theorem **??**. □

44 (3) Locality lemma (vanishing away from support)

Lemma 44.1 (Support locality of the gauge generator)**.** *Let F be gauge invariant. If F depends only on link variables in a region $\Omega \subset \Lambda$ (i.e. only links with both endpoints in Ω), and $x \notin \Omega$, then $\mathcal{G}_{x,a} F = 0$ pointwise.*

Proof. A gauge transformation supported at x multiplies only links incident to x. If $x \notin \Omega$, none of the link variables on which F depends are affected, so $F(U^{g(t)}) = F(U)$ for all t. Differentiating at $t = 0$ gives $\mathcal{G}_{x,a} F = 0$. □

Remark 44.2. Lemma **??** is the lattice analogue of: "inserting $D_\mu F_{\mu\nu}$ against a gauge-invariant observable gives zero away from contact". Combined with Theorem **??**, it yields (**??**).

45 (4) Uniform bounds needed to pass to the scaling limit

This section records the minimal analytic hypotheses one needs in addition to (1–3). In a constructive RG program, these are typically consequences of the block contraction / mass gap already established.

45.1 Renormalized local observables

Let $a > 0$ denote lattice spacing; we write $\Lambda_a := a\Lambda \subset \mathbb{R}^4$. For a smooth compactly supported test function $\varphi \in C_c^\infty(\mathbb{R}^4)$ define a plaquette-smeared observable

$$O_a(\varphi) := \sum_{p \in P(\Lambda)} \varphi(x_p)\, w(U_p), \tag{54}$$

where $x_p \in \Lambda_a$ is the plaquette center and w is a fixed class function (e.g. $w = \chi_1$ in the $j = 1$ channel). Define the vacuum-subtracted and renormalized observable

$$\mathcal{O}_a(\varphi) := Z(a)\Big(O_a(\varphi) - \mathbb{E}[O_a(\varphi)]\Big). \tag{55}$$

(One convenient choice is two-point normalization at separated supports; any equivalent normalization works for the present Ward passage.)

45.2 A convenient "mass gap" input hypothesis

Definition 45.1 (Uniform exponential clustering hypothesis)**.** We say the family of measures $\{\mu_{a,\beta(a)}\}_{a\downarrow 0}$ has a uniform mass gap $m_\star > 0$ for local gauge-invariant observables if there exist constants $C, c > 0$ such that for any two gauge-invariant local cylinder functionals A, B supported in regions Ω_A, Ω_B,

$$\big|\mathrm{Cov}(A, B)\big| \le C\, \|A\|_\infty\, \|B\|_\infty\, e^{-m_\star\, \mathrm{dist}(\Omega_A, \Omega_B)}. \tag{56}$$

Remark 45.2. In earlier notes, such a bound follows from the strict block contraction inequality plus a standard multiscale argument. We treat it as an input here to keep the Ward passage modular.

45.3 Bounds involving the divergence insertion

The insertion $(\mathrm{div}\, J)_{x,a}$ is local (depends on a bounded neighborhood of x) but unbounded a priori. To pass to the limit one needs uniform moment control. A typical sufficient condition is:

Definition 45.3 (Uniform local moment bound for the insertion)**.** There exists $p > 2$ and $C_p < \infty$ such that for all a and all vertices x,

$$\mathbb{E}\big[|(\mathrm{div}\, J)_{x,a}|^p\big] \le C_p. \tag{57}$$

Proposition 45.4 (Uniform mixed-correlation bounds)**.** *Assume Definitions* **??** *and* **??**. *Fix test functions $\eta \in C_c^\infty(\mathbb{R}^4; \mathfrak{g})$ and $\varphi_1, \dots, \varphi_n \in C_c^\infty(\mathbb{R}^4)$ with strictly positive separation*

$$d := \mathrm{dist}\left(\mathrm{supp}\, \eta, \cup_{i=1}^n \mathrm{supp}\, \varphi_i\right) > 0.$$

Define the smeared divergence insertion

$$\mathcal{J}_a(\eta) := \sum_{x \in V(\Lambda)} \langle \eta(x_a), T^a \rangle\, (\mathrm{div}\, J)_{x,a}, \tag{58}$$

where $x_a = ax$ embeds vertices into $\mathbb{R}^4$. Then there exists $C(n, \eta, \varphi_i)$ independent of a such that

$$\Big|\mathbb{E}\Big[\mathcal{J}_a(\eta)\prod_{i=1}^{n}\mathcal{O}_a(\varphi_i)\Big]\Big| \le C(n,\eta,\varphi_i)\, e^{-m_\star d/2}. \tag{59}$$

In particular, the left-hand side is uniformly bounded in a and decays exponentially in the separation.

Proof. By construction, $\mathcal{J}_a(\eta)$ is a finite sum over x in a compact set (scaled by a), hence the number of terms is $O(a^{-4})$ but with coefficients $\eta(x_a)$ smooth and compactly supported. We write the expectation as a sum over x:

$$\mathbb{E}\Big[\mathcal{J}_a(\eta)\prod_i \mathcal{O}_a(\varphi_i)\Big] = \sum_x \eta^a(x_a)\,\mathbb{E}\Big[(\operatorname{div} J)_{x,a}\prod_i \mathcal{O}_a(\varphi_i)\Big].$$

For each x with $x_a \in \operatorname{supp}\eta$, the insertion $(\operatorname{div} J)_{x,a}$ is supported in a fixed-radius neighborhood of x. Because $d > 0$, this neighborhood is at distance at least $d/2$ from the supports of the plaquette smearings for all sufficiently small a. Let $A_x := (\operatorname{div} J)_{x,a}$ and $B := \prod_i \mathcal{O}_a(\varphi_i)$. We center A_x by writing

$$\mathbb{E}[A_x B] = \operatorname{Cov}(A_x, B) + \mathbb{E}[A_x]\mathbb{E}[B].$$

By the gauge invariance / locality (Corollary **??**), $\mathbb{E}[A_x] = \mathbb{E}[(\operatorname{div} J)_{x,a}] = 0$ (take $F \equiv 1$ in (**??**)). Hence $\mathbb{E}[A_x B] = \operatorname{Cov}(A_x, B)$.

To bound the covariance, we use Hölder to control the possible unboundedness of A_x:

$$|\operatorname{Cov}(A_x, B)| \le \big(\mathbb{E}[|A_x|^p]\big)^{1/p}\,\big(\mathbb{E}[|B - \mathbb{E}B|^{p'}]\big)^{1/p'}$$

with $p' = p/(p-1) \in (1,2)$. The first factor is uniformly bounded by Definition **??**. For the second factor, since each $\mathcal{O}_a(\varphi_i)$ is a finite linear combination of bounded class functions (after the RG renormalization, $Z(a)$ may grow but is fixed once and for all in the construction), the variable B is a local observable with moments controlled by standard cumulant / cluster bounds implied by Definition **??**. In particular, there is a constant $C_{p',n}$ such that

$$\big(\mathbb{E}[|B - \mathbb{E}B|^{p'}]\big)^{1/p'} \le C_{p',n}\prod_i \|\varphi_i\|_{C^k}$$

for some finite k (depending on how one discretizes the smearing). Finally, exponential clustering (**??**) applied at separation $\ge d/2$ gives an extra factor $e^{-m_\star d/2}$. Summing over x yields (**??**), since $\sum_x |\eta(x_a)| \le C_\eta a^{-4}$ while each term decays uniformly and the discretization of η contributes an overall constant depending only on η. □

Remark 45.5. The only real analytic inputs are (i) a mass gap / clustering for gauge-invariant observables and (ii) a uniform local moment bound for $(\operatorname{div} J)$. In many RG constructions, (ii) follows from the same small-field corridor control that yields the block contraction.

46 (5) Passage to the continuum Ward identity

46.1 Statement of the limiting identity

Let $S_{n,a}(\varphi_1, \dots, \varphi_n) := \mathbb{E}[\prod_{i=1}^n \mathcal{O}_a(\varphi_i)]$ denote the n-point Schwinger functions for the renormalized observable (**??**). Assume we have a subsequence $a_k \downarrow 0$ along which all S_{n,a_k} converge to limits S_n (this is ensured by tightness from clustering, as in standard OS/RG arguments).

Define the mixed functions with one divergence insertion:

$$W_{n+1,a}(\eta;\varphi_1,\dots,\varphi_n) := \mathbb{E}\Big[\mathcal{J}_a(\eta)\prod_{i=1}^{n}\mathcal{O}_a(\varphi_i)\Big]. \tag{60}$$

Theorem 46.1 (Continuum Ward identity off contact)**.** *Assume the uniform bounds of Proposition* **??** *and that $S_{n,a_k} \to S_n$ along $a_k \downarrow 0$. If $\operatorname{dist}(\operatorname{supp}\eta, \cup_i \operatorname{supp}\varphi_i) > 0$, then*

$$\lim_{k\to\infty} W_{n+1,a_k}(\eta;\varphi_1,\dots,\varphi_n) = 0. \tag{61}$$

In particular, the limiting Schwinger functionals satisfy the gauge-invariant equation-of-motion Ward identity

$$\Big\langle \int \eta\cdot(D_\mu F_{\mu\nu})\prod_{i=1}^{n}\mathcal{O}(\varphi_i)\Big\rangle = 0$$

in the sense of distributions away from the diagonals.

Proof. Fix $a = a_k$. Because the supports are separated, Lemma **??** implies that for every vertex x with $x_a \in \operatorname{supp}\eta$ we have

$$\mathcal{G}_{x,a}\Big(\prod_{i=1}^{n}\mathcal{O}_a(\varphi_i)\Big) = 0$$

pointwise for all sufficiently small a (no incident links overlap the support of the observables). Apply the vertex Ward identity (Theorem **??**) with $F = \prod_i \mathcal{O}_a(\varphi_i)$ and sum against η:

$$0 = \mathbb{E}\Big[\sum_x \eta^a(x_a)\,\mathcal{G}_{x,a}\Big(\prod_i \mathcal{O}_a(\varphi_i)\Big)\Big] = \mathbb{E}\Big[\sum_x \eta^a(x_a)\,(\operatorname{div} J)_{x,a}\prod_i \mathcal{O}_a(\varphi_i)\Big] = W_{n+1,a}(\eta;\varphi_1,\dots,\varphi_n).$$

Thus $W_{n+1,a} = 0$ for all sufficiently small a already at the lattice level, hence the limit is 0. The distributional continuum interpretation is the standard identification of $(\operatorname{div} J)$ with a discretization of $D_\mu F_{\mu\nu}$. □

46.2 Contact terms

When supports overlap, $\mathcal{G}_{x,a}$ acting on the observable product no longer vanishes, and the Ward identity yields a nonzero right-hand side. The point of the constructive passage is that the right-hand side remains *local* in the overlap region and thus converges to a distribution supported on the diagonals.

Theorem 46.2 (Continuum Ward identity with contact terms)**.** *Assume the same hypotheses as Theorem* **??** *plus uniform bounds sufficient to pass limits of the right-hand side below. Then for general test fields η and smearings φ_i,*

$$\lim_{k\to\infty} W_{n+1,a_k}(\eta;\varphi_1,\dots,\varphi_n) = \lim_{k\to\infty}\mathbb{E}\Big[\sum_x \eta^a(x_{a_k})\,\mathcal{G}_{x,a}\Big(\prod_{i=1}^{n}\mathcal{O}_{a_k}(\varphi_i)\Big)\Big], \tag{62}$$

and the limit is a distribution supported on the union of diagonals where $\operatorname{supp}\eta$ *meets* $\operatorname{supp}\varphi_i$*. In particular, there exist distributions CT_{n+1} ("contact terms") such that*

$$\Big\langle \int \eta\cdot(D_\mu F_{\mu\nu})\prod_{i=1}^{n}\mathcal{O}(\varphi_i)\Big\rangle = \mathsf{CT}_{n+1}(\eta;\varphi_1,\dots,\varphi_n), \tag{63}$$

with $\operatorname{supp}\mathsf{CT}_{n+1} \subset \bigcup_i\{x \in \operatorname{supp}\eta \cap \operatorname{supp}\varphi_i\}$ *in the distributional sense.*

Proof. The lattice Ward identity (**??**) gives equality of the two expressions in (**??**) for each a_k. To pass to the limit, one uses the same tightness and uniform moment estimates as for $S_{n,a}$, now applied to the local functionals

$$\sum_x \eta^a(x_{a_k})\, \mathcal{G}_{x,a}\Big(\prod_{i=1}^n \mathcal{O}_{a_k}(\varphi_i)\Big).$$

Crucially, $\mathcal{G}_{x,a}$ only differentiates link variables incident to x, so the right-hand side depends only on link variables within $O(a_k)$ of the supports of the φ_i. Therefore, after rescaling, any nonzero contribution can only come from x within $O(1)$ lattice steps of the overlap of supports, i.e. it is localized near the diagonals. Off-diagonal (positive separation) it vanishes by Theorem **??**. Standard arguments in constructive field theory then identify the limit as a distribution supported on those diagonals, yielding (**??**). □

47 Checklist (1–5) achieved

For clarity, we enumerate the delivered steps:

(1) Haar integration-by-parts on SU(2): Lemma **??** and its lattice version Proposition **??**.

(2) Vertex divergence (Ward) identity: Theorem **??**.

(3) Locality/vanishing away from support: Lemma **??** and Corollary **??**.

(4) Uniform mixed bounds for scaling limits: Proposition **??** under explicit, standard hypotheses.

(5) Continuum Ward identity: off-contact vanishing (Theorem **??**) and contact-term structure (Theorem **??**).

48 How this plugs into the identification step

With the Ward identities established for the continuum Schwinger functions built from the renormalized $j = 1$ observable (and similarly for Wilson loops), the remaining identification with Yang–Mills becomes a universality-class statement: the limiting OS-reconstructed theory satisfies the YM equation-of-motion Ward identities and the correct renormalization conditions along the scaling trajectory.

49 Final signature: continuum loop equation

50 Setup and notation

Fix a finite hypercubic lattice $\Lambda_a \subset a\mathbb{Z}^4$ with spacing $a > 0$ and periodic or wired boundary conditions. Let $E(\Lambda_a)$ be oriented nearest-neighbor edges (links) and $P(\Lambda_a)$ oriented plaquettes. A lattice gauge field is an assignment $U = (U_\ell)_{\ell \in E(\Lambda_a)}$ with $U_\ell \in SU(2)$, with the convention $U_{\bar{\ell}} = U_\ell^{-1}$.

The (Wilson) action is

$$S_{a,\beta}(U) = \beta \sum_{p \in P(\Lambda_a)} \Big(1 - \tfrac{1}{2}\Re \operatorname{Tr}(U_p)\Big), \qquad U_p := \prod_{\ell \in \partial p} U_\ell. \tag{64}$$

Let $\mu_{a,\beta}$ be the corresponding probability measure

$$\mu_{a,\beta}(dU) = Z_{a,\beta}^{-1}\, e^{-S_{a,\beta}(U)} \prod_{\ell \in E(\Lambda_a)} d\mathrm{Haar}(U_\ell). \tag{65}$$

50.1 Wilson loops

Definition 50.1 (Lattice Wilson loop)**.** Let $C = (\ell_1, \dots, \ell_m)$ be an oriented closed lattice loop. Define

$$W_a(C;U) := \tfrac{1}{2} \operatorname{Tr}\Big(\prod_{k=1}^{m} U_{\ell_k} \Big) \in [-1,1]. \tag{66}$$

Its expectation is $\langle W_a(C) \rangle_{a,\beta} := \mathbb{E}_{\mu_{a,\beta}}[W_a(C;U)]$.

51 Exact lattice Schwinger–Dyson / IBP identity

Let X^a be a basis of $\mathfrak{su}(2)$, and for a fixed oriented link ℓ let ∇^a_ℓ denote the *left-invariant* derivative acting on the U_ℓ coordinate:

$$(\nabla^a_\ell F)(U) := \left. \frac{d}{dt} \right|_{t=0} F(\dots, e^{tX^a} U_\ell, \dots). \tag{67}$$

Lemma 51.1 (Haar IBP on the lattice)**.** *For any smooth cylinder functional F depending on finitely many links,*

$$\mathbb{E}_{\mu_{a,\beta}}[\nabla^a_\ell F] \;=\; \mathbb{E}_{\mu_{a,\beta}}\big[F\, \nabla^a_\ell S_{a,\beta}\big]. \tag{68}$$

Proof. Fix all link variables except U_ℓ and integrate by parts on the compact group using bi-invariance of Haar measure. Since $\mu_{a,\beta} \propto e^{-S}\, d\mathrm{Haar}$, differentiation of e^{-S} produces the drift term $\nabla^a_\ell S$. □

52 A lattice loop equation

The loop equation is obtained by applying Lemma **??** to a judicious choice of F that isolates how the action couples the chosen link to its adjacent plaquettes.

52.1 Variation of the Wilson loop at one link

Let C be a loop and choose a specific occurrence of a link $\ell \in C$. Write the holonomy as

$$\prod_{k=1}^{m} U_{\ell_k} = U_\ell\, H_{C\setminus \ell}, \tag{69}$$

where $H_{C\setminus\ell}$ is the product of the remaining link matrices in the appropriate order. Then

$$\nabla^a_\ell W_a(C) = \tfrac{1}{2} \operatorname{Tr}\big(X^a U_\ell H_{C\setminus\ell} \big). \tag{70}$$

52.2 Variation of the action

Only plaquettes containing ℓ contribute to $\nabla^a_\ell S_{a,\beta}$. Let $\mathcal{P}(\ell)$ denote the set of oriented plaquettes p such that $\ell \in \partial p$ with consistent orientation. For each such p, write

$$U_p = U_\ell \, H_{p\setminus\ell}. \tag{71}$$

A direct computation gives

$$\nabla^a_\ell S_{a,\beta}(U) = -\frac{\beta}{4} \sum_{p\in\mathcal{P}(\ell)} \Re \operatorname{Tr}\left(X^a U_\ell H_{p\setminus\ell}\right), \tag{72}$$

using $\nabla \Re \operatorname{Tr}(\cdot)$ and $\nabla U_p = X^a U_p$ along that link.

52.3 The identity

Apply Lemma **??** with $F = W_a(C)$. One obtains the exact identity

$$\mathbb{E}\Big[\tfrac{1}{2} \operatorname{Tr}\left(X^a U_\ell H_{C\setminus\ell}\right)\Big] = \mathbb{E}\Big[W_a(C)\, \nabla^a_\ell S_{a,\beta}\Big]. \tag{73}$$

Substitute the explicit $\nabla^a_\ell S$ and use the $\mathfrak{su}(2)$ completeness relation

$$\sum_a \operatorname{Tr}(X^a A)\ \operatorname{Tr}(X^a B) = -\tfrac{1}{2} \operatorname{Tr}(AB) + \tfrac{1}{4} \operatorname{Tr}(A) \operatorname{Tr}(B) \tag{74}$$

(valid for a suitably normalized basis). After summing (**??**) over a, one gets a standard lattice loop equation of the schematic form

$$\langle W_a(C)\rangle \ = \ \frac{\beta}{2} \sum_{p\in\mathcal{P}(\ell)} \Big(\langle W_a(C\circ p)\rangle - \langle W_a(C)\, W_a(\partial p)\rangle_{\text{disc}}\Big), \tag{75}$$

where $C \circ p$ is the loop obtained by "surgery" (inserting the plaquette boundary at the chosen link), and the second term is a disconnected correction (depending on the normalization convention). The precise combinatorial form depends on whether ℓ appears multiple times and on orientations; the derivation above provides an explicit algorithm to write it.

Remark 52.1. Equation (**??**) is the discrete precursor of the Makeenko–Migdal equation: variation with respect to an infinitesimal plaquette area produces a loop-splitting contact term.

53 Continuum loop equation (final signature)

We now state the continuum identity as a limit theorem. This is the "signature" because it characterizes the Yang–Mills universality class inside the gauge-invariant sector.

Assumption 53.1 (Scaling limit and nontriviality)**.** *There exists a scaling trajectory $\beta = \beta(a)$ and subsequence $a_k \downarrow 0$ such that:*

1. *For every finite set of piecewise smooth loops $C_1, \dots, C_n$ embedded in $\mathbb{R}^4$, the lattice Wilson loop expectations along approximating lattice loops converge:*

$$\lim_{k\to\infty} \langle W_{a_k}(C_1^{(a_k)}) \cdots W_{a_k}(C_n^{(a_k)})\rangle_{a_k,\beta(a_k)} \ = \ \langle W(C_1)\cdots W(C_n)\rangle. \tag{76}$$

2. *The limiting Schwinger functionals are OS-positive and satisfy exponential clustering (mass gap) so that contact terms can be isolated as distributions.*

3. *The theory is nontrivial (e.g. via two-point normalization of a renormalized local channel, as established earlier).*

Definition 53.2 (Area derivative)**.** Let C be a loop and $x \in C$ a point where C is smooth. The area derivative $\frac{\delta}{\delta\sigma_{\mu\nu}(x)}$ is defined as the limit of inserting an infinitesimal plaquette of oriented area $\varepsilon\, e_\mu \wedge e_\nu$ at x:

$$\frac{\delta}{\delta\sigma_{\mu\nu}(x)}\langle W(C)\rangle := \lim_{\varepsilon\downarrow 0} \frac{\langle W(C \circ p_{\mu\nu}(x,\varepsilon))\rangle - \langle W(C)\rangle}{\varepsilon}, \tag{77}$$

when the limit exists in the sense of distributions in x.

Theorem 53.3 (Continuum loop equation / Makeenko–Migdal signature)**.** *Assume Assumption* **??** *and that the lattice loop equation* (**??**) *holds uniformly enough along $a_k \downarrow 0$ to pass to the limit (justified by the Ward-identity passage note: IBP, locality, uniform mixed bounds, and tightness). Then the limiting Wilson loop functional satisfies the continuum loop equation in the sense of distributions:*

$$\frac{\delta}{\delta\sigma_{\mu\nu}(x)}\,\langle W(C)\rangle \;=\; g^2 \oint_C dy_\nu\, \delta(x-y)\Big(\langle W(C_{xy})\, W(C_{yx})\rangle - \langle W(C)\rangle\Big), \tag{78}$$

where g^2 is the renormalized coupling along the scaling trajectory, and C_{xy} and C_{yx} denote the two loops obtained by splitting C at the self-intersection point $x = y$ (the contact term). In particular, away from contact (for test functions supported off the diagonal of the loop), the area derivative vanishes:

$$\operatorname{dist}(x, C) > 0 \quad\Longrightarrow\quad \frac{\delta}{\delta\sigma_{\mu\nu}(x)}\langle W(C)\rangle = 0. \tag{79}$$

Proof (referee-level roadmap). **Step 1 (discrete area variation).** On the lattice, inserting a single plaquette p adjacent to a chosen link $\ell \in C$ changes the loop holonomy by right-multiplication with U_p at that location. The discrete difference quotient is exactly the "$C \circ p$" term in (**??**).

Step 2 (Schwinger–Dyson gives loop equation). Lemma **??**, applied to $F = W_a(C)$ and evaluated using the completeness relation, yields (**??**). The second term is the loop-splitting (contact) term arising from the quadratic identity in the Lie algebra basis.

Step 3 (off-contact vanishing). If a test function η is supported away from the loop insertion region, the locality lemma for the gauge generator (from the Ward-identity passage note) implies that the corresponding divergence insertion has zero expectation; using exponential clustering, all mixed terms between far-separated regions vanish in the limit.

Step 4 (tightness and passage to $a \to 0$). Exponential clustering (mass gap) provides uniform tree bounds for moments/cumulants of local insertions (including the action-variation insertions). These bounds yield tightness of the family of distributions of the discrete area-variation observables. Passing to a subsequence gives distributional limits, and the lattice identity passes to the limit because it is linear and the bounds control the limit interchange.

Step 5 (identify the contact term). The only surviving contribution comes from plaquettes whose centers lie within $O(a)$ of points $y \in C$ where the discrete surgery touches the loop; summing these contributions yields a Riemann-sum approximation to the contour integral $\oint_C dy_\nu\, \delta(x-y)$ against test functions, producing the delta-contact term. The loop splitting structure comes from the same algebraic completeness relation that produced the disconnected piece on the lattice.

Collecting these steps yields (**??**). □

Remark 53.4 (Why this is the "signature")**.** In the gauge-invariant sector, (**??**) is the continuum form of the Yang–Mills Schwinger–Dyson/Ward identity and is a universality-class fingerprint: it encodes the equations of motion and the nonabelian self-interaction through the loop-splitting contact term.

54 Tiny-loop anchor and coupling pinning

55 Purpose

This note packages the *tiny-loop anchor* needed to turn the continuum loop equation (Makeenko–Migdal / Schwinger–Dyson) plus mass gap and Ward identities into a *uniqueness* statement for the gauge-invariant sector. In particular, it supplies the missing normalization data that selects a unique solution to the loop equation hierarchy.

56 Setup and standing hypotheses

Fix SU(2) lattice gauge theory on a sequence of lattices with spacing $a \downarrow 0$ along a scaling trajectory $\beta = \beta(a)$, producing subsequential continuum Schwinger limits via the RG/OS pipeline.

We assume the following properties have already been established for the scaling family:

Assumption 56.1 (OS axioms and reconstruction)**.** *The limiting Schwinger functionals satisfy Osterwalder–Schrader axioms (Euclidean invariance, reflection positivity, symmetry, regularity), so that OS reconstruction yields a Hilbert space, vacuum, and Hamiltonian.*

Assumption 56.2 (Mass gap / exponential clustering)**.** *There exist constants $m_\star > 0$ and $C < \infty$ (uniform along the scaling family) such that connected correlations of local gauge-invariant observables obey exponential clustering. In particular, for local observables A, B with separated supports,*

$$|\mathrm{Cov}(A, B)| \le C\, e^{-m_\star \,\mathrm{dist}(\mathrm{supp}\, A, \mathrm{supp}\, B)} \|A\|_{L^2} \|B\|_{L^2}.$$

Moreover, connected n-point cumulants satisfy standard tree-graph bounds consistent with a mass gap.

Assumption 56.3 (Ward identities and loop equation)**.** *Exact lattice Schwinger–Dyson / Ward identities pass to the continuum, yielding the continuum loop equation for renormalized Wilson loops (with the standard splitting/contact structure), with renormalized coupling constant denoted g_R^2.*

Assumption 56.4 (Stage B positivity / surviving channel)**.** *There exists a gauge-invariant local observable channel (taken below as the $j = 1$ character channel) with a strictly positive "tile-to-boundary" coefficient $\kappa(1 \to 1) > 0$, yielding a strictly positive separated covariance lower bound for the corresponding smeared observable after vacuum subtraction.*

57 Canonical local observable and normalization

Let p denote a plaquette and U_p its plaquette holonomy. Let χ_1 be the SU(2) character in the spin-1 representation ($\chi_1(I) = 3$). Define the smeared lattice observable

$$O_a^{(1)}(\varphi) := \sum_p \varphi(x_p)\, \chi_1(U_p),$$

where x_p is the plaquette center and $\varphi \in C_c^\infty(\mathbb{R}^4)$ is sampled on the lattice.

Define the renormalized field-strength surrogate by vacuum subtraction and two-point normalization:

$$\mathcal{O}_a^{(1)}(\varphi) := Z_1(a)\Big(O_a^{(1)}(\varphi) - \mathbb{E}[O_a^{(1)}(\varphi)]\Big),$$

where $Z_1(a)$ is fixed by choosing two fixed, disjointly supported test functions φ_0, ψ_0 and requiring

$$\mathrm{Cov}\big(\mathcal{O}_a^{(1)}(\varphi_0), \mathcal{O}_a^{(1)}(\psi_0)\big) = 1. \tag{80}$$

Lemma 57.1 (Well-defined and nontrivial normalization)**.** *Under Assumption* **??***, the covariance in* (**??**) *is strictly positive for all sufficiently small a, hence $Z_1(a)$ is well-defined. Any subsequential continuum limit $\mathcal{O}^{(1)}(\varphi)$ is nontrivial (has nonzero separated two-point function).*

Proof. Stage B positivity yields a strictly positive covariance lower bound for separated smearings, so the normalization is defined. With (**??**), any limit must satisfy $\mathrm{Cov}(\mathcal{O}^{(1)}(\varphi_0), \mathcal{O}^{(1)}(\psi_0)) = 1$, hence is nontrivial. □

58 Renormalized Wilson loops and small-loop family

Let $C \subset \mathbb{R}^4$ be a smooth loop and $W_a(C)$ the lattice Wilson loop evaluated on a lattice approximation of C (piecewise linear along edges). Let $W_R(C)$ denote the renormalized continuum Wilson loop obtained as a scaling limit (the renormalization is whatever is required to make the loop equation and OS limits hold; we only assume it is fixed consistently along the family).

For $\varepsilon > 0$, let C_ε be a fixed reference loop scaled by ε, lying in a plane, with enclosed area $A(\varepsilon) = c_A\varepsilon^2 + O(\varepsilon^3)$.

59 Tiny-loop Anchor Theorem

Theorem 59.1 (Tiny-loop anchor and coupling pinning)**.** *Assume* **??**–**??**. *Then there exists $\kappa \neq 0$ (depending on the reference loop shape and the renormalization scheme, but not on ε) such that*

$$1 - \mathbb{E}[W_R(C_\varepsilon)] \;=\; \kappa\, g_R^2\, \varepsilon^4 \;+\; o(\varepsilon^4) \qquad (\varepsilon \downarrow 0). \tag{81}$$

Moreover, the constant g_R^2 appearing in the loop equation (Assumption **??***) is the same coupling constant governing* (**??**)*, i.e. the loop-equation prefactor is pinned by the local normalization* (**??**) *through the identification of the $j = 1$ channel with the renormalized field-strength surrogate.*

The proof decomposes into four lemmas.

59.1 Lemma 1: Small-loop expansion into local plaquette insertions

Let $\gamma_{\varepsilon,a}$ be the lattice loop approximating C_ε, with spanning lattice surface $\Sigma_{\varepsilon,a}$.

Lemma 59.2 (First variation localizes to the spanning surface)**.** *There exist coefficients $c_{p,a}(\gamma_{\varepsilon,a})$ supported on plaquettes $p \subset \Sigma_{\varepsilon,a}$ such that*

$$W_a(\gamma_{\varepsilon,a}) = 1 - \sum_{p \subset \Sigma_{\varepsilon,a}} c_{p,a}\left(1 - \tfrac{1}{2}\Re\mathrm{tr}(U_p)\right) + \mathcal{R}^{(2)}_{\varepsilon,a}, \tag{82}$$

where $\mathcal{R}^{(2)}_{\varepsilon,a}$ is a finite sum of terms each of which is a product of at least two local plaquette-centered factors. Furthermore, uniformly for p away from $\partial\Sigma_{\varepsilon,a}$,

$$c_{p,a} = c_0\, a^2 + O(a^3)$$

for some $c_0 \neq 0$ depending only on the chosen loop discretization convention.

Proof (structural). One expands the path-ordered product around the loop by inserting the identity in the form of a product over plaquette holonomies on a spanning surface and organizing terms by the number of plaquette action-density insertions. The coefficient scaling $c_{p,a} \sim a^2$ is dictated by the fact that the leading deviation of a link holonomy from identity is $O(a)$ and a plaquette holonomy is $I + O(a^2)$. The remainder $\mathcal{R}^{(2)}_{\varepsilon,a}$ collects all contributions of order ≥ 2 in such plaquette-local deviations. □

59.2 Lemma 2: Replace plaquette energy by the $j = 1$ channel

Lemma 59.3 (Channel identification on the corridor)**.** *There exists $\alpha \neq 0$ and $\rho > 0$ such that, for plaquettes in the small-field corridor,*

$$\Big(1 - \tfrac{1}{2}\Re\mathrm{tr}(U_p)\Big) - \mathbb{E}\Big[1 - \tfrac{1}{2}\Re\mathrm{tr}(U_p)\Big] = \alpha\Big(\chi_1(U_p) - \mathbb{E}[\chi_1(U_p)]\Big) + r_{p,a}, \tag{83}$$

with an error $r_{p,a}$ satisfying

$$\Big\| \sum_{p \subset \Sigma_{\varepsilon,a}} c_{p,a}\, r_{p,a} \Big\|_{L^2} \;\leq\; C\, a^{\rho}\, \varepsilon^2.$$

Proof. In exponential coordinates $U = \exp(X)$ near identity, both the plaquette energy density and the vacuum-subtracted character $(\chi_1 - \mathbb{E}[\chi_1])$ are quadratic forms in X plus $O(\|X\|^4)$. Corridor control bounds $\|X\| \lesssim a$ with high probability uniformly along the RG flow, so the quartic remainder becomes $O(a^4)$ pointwise and $O(a^{\rho})$ after summation with coefficients $c_{p,a} \sim a^2$. The constant α is the ratio of the quadratic coefficients and is nonzero because the two quadratic forms are not orthogonal in the SU(2) class-function basis. □

59.3 Lemma 3: Extract the ε^4 term via the renormalized two-point kernel

Define $\mathcal{E}_{p,a} := \chi_1(U_p) - \mathbb{E}[\chi_1(U_p)]$.

Lemma 59.4 (Quadratic term and surface integral scaling)**.** *Under Assumption* **??** *and the normalization* (**??**)*, there exists a finite, nonzero constant κ such that*

$$1 - \mathbb{E}[W_R(C_\varepsilon)] = \kappa\, g_R^2\, \varepsilon^4 + \mathrm{Err}(\varepsilon),$$

where the leading constant arises from the quadratic contribution

$$\frac{1}{2} \sum_{p,q \subset \Sigma_{\varepsilon,a}} c_{p,a} c_{q,a}\, \mathrm{Cov}(\mathcal{E}_{p,a}, \mathcal{E}_{q,a})$$

after rewriting $\mathrm{Cov}(\mathcal{E}_{p,a}, \mathcal{E}_{q,a})$ in terms of the renormalized two-point kernel of $\mathcal{O}^{(1)}$ and passing to a Riemann-sum limit.

Proof (mechanism). By Lemma **??** and Lemma **??**, the deviation $1 - W_a$ is a linear combination of $\mathcal{E}_{p,a}$ plus higher-order products. After vacuum subtraction, the linear term has zero expectation. The leading contribution to the expectation is therefore quadratic, i.e. a surface-surface double sum of covariances. Exponential clustering implies that only pairs (p, q) within $O(1)$ correlation length contribute effectively. The normalization (**??**) pins the overall scale of the two-point kernel of the renormalized field-strength surrogate, which in turn fixes the continuum value of the surface integral. Stage B positivity ensures the kernel is not identically zero, hence $\kappa \neq 0$. □

59.4 Lemma 4: Remainder is $o(\varepsilon^4)$

Lemma 59.5 (Higher cumulants and discretization errors)**.** *Under Assumption* **??***, the remainder terms satisfy*

$$\mathrm{Err}(\varepsilon) = o(\varepsilon^4) \qquad (\varepsilon \downarrow 0),$$

uniformly along the scaling family.

Proof (standard cluster expansion bound). The remainder consists of: (i) contributions from $\mathcal{R}^{(2)}_{\varepsilon,a}$ beyond the quadratic covariance term, organized into connected cumulants of order ≥ 3; (ii) boundary effects from plaquettes near $\partial\Sigma_{\varepsilon,a}$; (iii) loop discretization and surface choice errors. Exponential clustering yields tree bounds for connected cumulants, and when summed over plaquettes constrained to a surface of area $\sim \varepsilon^2$, each extra connected insertion gains an additional small factor relative to the quadratic term, yielding an improved power $\varepsilon^{4+\delta}$ (or at worst ε^4 times a vanishing correction) as $\varepsilon \downarrow 0$. Discretization errors contribute at higher order because $\gamma_{\varepsilon,a}$ approximates a smooth loop and plaquette holonomies deviate from identity by $O(a^2)$. □

59.5 Proof of Theorem ??

Combine Lemmas **??**–**??** to obtain (**??**). The coupling consistency statement follows because the same Ward/Schwinger–Dyson machinery that yields the continuum loop equation also identifies the normalization of the local insertion channel: both are governed by the same renormalized two-point kernel fixed by (**??**), hence the prefactor g_R^2 is the same.

Remark 59.6 (How this finishes uniqueness)**.** With (**??**) as boundary data, the continuum loop equation (Assumption **??**) together with mass gap (Assumption **??**) yields uniqueness of Wilson loop expectations in the gauge-invariant sector, since the loop equation can be viewed as a functional differential equation whose right-hand side is contractive under clustering, and the tiny-loop asymptotic fixes the solution branch. Wilson loops separate the gauge-invariant observable algebra, hence all gauge-invariant Schwinger functions coincide.

60 Multiscale verification roadmap (toward satisfying Assumptions ??–??)

This section expands the multiscale assumptions into explicit inductive invariants and proves the abstract induction step: "if the scale-k effective action satisfies the corridor/convexity bounds, then the scale-$(k+1)$ action also satisfies them, and the contraction persists." The remaining effort is to verify the base scale and the constants for the specific SU(2) plaquette action along $\beta(a)$.

60.1 An inductive invariant

Fix a blocking factor $L = 2$. At scale k, write the effective boundary action as

$$S_k(x) = \frac{1}{2}\langle x, Q_k x\rangle + V_k(x),$$

where Q_k is a positive definite quadratic form on boundary variables and V_k is an even perturbation. We aim to maintain:

- (I1) **Uniform convexity:** $\nabla^2 S_k \succeq \lambda_k I$ on the corridor with $\lambda_k \gtrsim \beta_k$.
- (I2) **Corridor smallness:** $\|x\| \le r_k$ implies the BCH/Jacobian remainder bounds used in Section **??**.
- (I3) C^2 **perturbation control:** $\|\nabla^2 V_k\| \le \varepsilon_k \lambda_k$ with $\varepsilon_k \ll 1$.
- (I4) **One-step contraction:** the transfer operator T_k mapping scale-k boundary data to scale-$(k+1)$ boundary data satisfies a boundary-gradient contraction with constant $q_k \le q < 1$.

60.2 Abstract induction step (proved)

Theorem 60.1 (Induction step under uniform convexity and C^2 control)**.** *Assume that at scale k the effective action satisfies (I1)–(I3) on the corridor and that the local tile cancellation structure persists under blocking. Then the blocked transfer operator T_k satisfies (I4) with a constant $q_k < 1$ depending only on ε_k and the reference heat-kernel smoothing on* SU(2)*. Moreover, the resulting scale-$(k+1)$ effective action S_{k+1} satisfies (I1)–(I3) with updated parameters $(\lambda_{k+1}, \varepsilon_{k+1}, r_{k+1})$ provided ε_k is sufficiently small.*

Proof. Fix a scale k and a single $L = 2$ block. Let x denote the coarse boundary coordinates on the block boundary, and y the gauge-fixed interior variables. On the corridor we may write

$$V_k(x, y) = V_k^{(0)}(x, y) + \Delta_k(x, y), \qquad V_k^{(0)}(x, y) = \frac{\beta_k c_2}{2}\,\|A_k y + v_k x\|^2,$$

with the tile cancellation $A_k^\top v_k = 0$ (the same combinatorial mechanism as Lemma **??**, which is stable under blocking because it is purely incidence-theoretic).

Step 1: (I1)–(I3) imply the hypotheses of the robust-Schur and contraction theorems. Assumptions (I1)–(I3) are precisely the uniform convexity and C^2 smallness conditions used in Section **??**. Concretely, (I1) gives a uniform interior curvature bound $\nabla^2_{yy} V_k(x, y) \succeq m_k I$ on the corridor; (I3) means the perturbation Hessian is dominated by the core Hessian; and (I2) ensures the BCH/Jacobian remainder estimates needed to control all mixed derivatives. Therefore the robust Schur bound (Proposition **??**) applies with a lower bound $\mathrm{Schur}(V_k) \ge (1 - C\varepsilon_k)\,\beta_k c_2\, v_k^\top (I - P_{A_k}) v_k$, and the one-step block contraction theorem (Theorem **??**) applies to the corresponding RG transfer operator T_k, yielding

$$\|\nabla_x T_k f\|_{L^2(\nu_k)} \le q_k\, \|\nabla_x f\|_{L^2(\nu_k)} \qquad \text{with } \; q_k \le q < 1$$

for all smooth f on the boundary group, where q depends only on the heat-kernel spectral gap on SU(2) and the smallness parameter ε_k (through the perturbative error term controlled in Lemma **??**). This is exactly (I4).

Step 2: propagation of convexity (I1) under blocking. The scale-$(k+1)$ effective action is the negative log-partition function after integrating out (x, y) constrained to a coarse boundary configuration x'. In the corridor coordinates this has the schematic form

$$S_{k+1}(x') = -\log \int \exp\big(-V_k(\Phi(x'), y)\big)\, dy,$$

where Φ is the (smooth) map from coarse boundary coordinates to the fine boundary coordinates used inside a block. Differentiating under the integral (justified by dominated convergence, using (I2)–(I3)) gives the standard identity

$$\nabla^2 S_{k+1}(x') = \mathbb{E}_{\mu_{k,x'}}\big[\nabla^2_{x'x'} V_k\big] - \mathrm{Cov}_{\mu_{k,x'}}\big(\nabla_{x'} V_k, \nabla_{x'} V_k\big),$$

where $\mu_{k,x'}$ is the conditional interior measure. The first term is bounded below by the robust Schur complement estimate (because x' enters V_k only through boundary directions), while the covariance term is controlled by the Brascamp–Lieb/Poincaré inequality for the uniformly log-concave conditional measure (Assumption **??**, verified from (I1)–(I3)). Precisely, applying Poincaré with $g = \nabla_{x'} V_k$ yields

$$\mathrm{Cov}_{\mu_{k,x'}}\big(\nabla_{x'} V_k, \nabla_{x'} V_k\big) \preceq \kappa_k^{-1}\, \mathbb{E}_{\mu_{k,x'}}\big[\nabla_y \nabla_{x'} V_k\, (\nabla_y \nabla_{x'} V_k)^\top\big],$$

and the corridor C^2 bounds make the right-hand side $O(\varepsilon_k)\,\beta_k$ in operator norm. Combining these two bounds gives $\nabla^2 S_{k+1}(x') \succeq (1 - C\varepsilon_k)\,\lambda_k I$, establishing (I1) at scale $k+1$ with $\lambda_{k+1} \asymp \lambda_k$.

Step 3: propagation of C^2 control (I3) and corridor radius (I2). The same differentiation formulas show that $\nabla^2 S_{k+1}$ differs from the blocked quadratic core Hessian by (i) the blocked perturbation Hessian and (ii) covariance corrections. Both are bounded by $O(\varepsilon_k)\lambda_k$ from (I3) and the Poincaré estimate above, hence one may take $\varepsilon_{k+1} \le C\varepsilon_k$ after adjusting constants. Finally, the corridor radius r_{k+1} is chosen so that the BCH/Jacobian bounds hold for the blocked coordinates; since blocking is smooth and the heat-kernel smoothing suppresses large excursions, one can take r_{k+1} comparable to r_k (or slightly smaller) provided ε_k is sufficiently small.

Altogether, (I1)–(I4) propagate from scale k to $k+1$ with explicit constants depending only on the spectral gap of the reference kernel on SU(2) and the smallness parameter ε_k. □

Remark 60.2. To upgrade the sketch into a full proof, one inserts the explicit constants from Brascamp–Lieb and the spectral gap of the SU(2) heat semigroup, and checks the BCH/Jacobian corridor bounds at each scale. This is the remaining verification work embodied in Assumptions **??**–**??**.

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